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4. NOISE

4.1 Introduction

4.1.1.1 This section presents an assessment on the potential noise impacts arising from the construction and operation of Northern Metropolis Highway – San Tin Section ("the Project"). The noise impact assessment is conducted in accordance with the requirements of Annexes 5 and 13 of the Technical Memorandum on Environmental Impact Assessment Process as well as the requirements set out under Clause 3.5.4 and Appendix C of the EIA Study Brief (No. ESB-373/2025).

4.2 Environmental Legislation, Standards and Criteria

4.2.1 General

4.2.1.1 Noise impact has been assessed in accordance with the criteria and methodology given in the Technical Memoranda (TM) under the Noise Control Ordinance (NCO) (Cap. 400), and the EIAO-TM under the Environmental Impact Assessment Ordinance (EIAO) (Cap. 499).

4.2.1.2 The NCO and EIAO provide the statutory framework for noise control. Assessment procedures and standards are set out in the following TMs and documents:

- Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM);
- Technical Memorandum for the Assessment of Noise from Places other than Domestic Premises, Public Places or Construction Sites (IND-TM);
- Technical Memorandum on Noise from Construction Work in Designated Areas (DA-TM);
- Technical Memorandum on Noise from Construction Work other than Percussive Piling (GW-TM);
- Technical Memorandum on Noise from Percussive Piling (PP-TM);
- EIAO Guidance Note No. 9/2023 – Preparation of Construction Noise Impact Assessment under the Environmental Impact Assessment Ordinance (GN No. 9/2023);
- EIAO Guidance Note No. 12/2023 – Preparation of Road Traffic Noise Impact Assessment under the Environmental Impact Assessment Ordinance (GN No. 12/2023);
- EIAO Guidance Note No. 16/2023 – Preparation of Fixed Noise Sources Impact Assessment under the Environmental Impact Assessment Ordinance (GN No. 16/2023);
- ETWB TCW 13/2003 – Guidelines and Procedures for Environmental Impact Assessment of Government Projects and Proposals;
- ETWB TCW 13/2003/A – Guidelines and Procedures for Environmental Impact Assessment of Government Projects and Proposals – Planning for Provision of Noise Barrier;
- Practice Note for Professional Persons on Minimizing Noise from Construction Activities (ProPECC PN 1/24); and

- Practice Note for Professional Persons on Application of Innovative Noise Mitigation Designs in Planning Private Residential Developments against Road Traffic Noise Impact (ProPECC PN 5/23).

4.2.2 Construction Noise

Construction Airborne Noise during Non-restricted Hours

4.2.2.1 Noise from construction activities taking place at 0700 – 1900 hours on any day not being a Sunday or general holiday is subject to the Noise Standards for Daytime Construction Activities stated in Table 1B of Annex 5 of the EIAO-TM. The noise criteria are reproduced in **Table 4.1**. For general construction work to be carried out during non-restricted hours, guidelines in GN No. 9/2023 should be followed.

Table 4.1 Noise Standards for Daytime Construction Activities

Uses	**Daytime Construction Airborne Noise Criteria, Leq (30 mins) dB(A)**
• All domestic premises; • Temporary housing accommodation; • Hostels; • Convalescent homes; and • Homes for the aged	75
• Places of public worship; • Courts of law; and • Hospitals and medical clinics	70
• Educational institutions (including kindergartens and nurseries)	70 65 (during examinations)
Notes: 1. The above standards apply to uses which rely on opened windows for ventilation and are assessed at 1m from the external façade.	

Construction Airborne Noise during Restricted Hours

4.2.2.2 The NCO also provides statutory control on general construction works during restricted hours (i.e. between 1900 and 0700 hours and any time on a general holiday, including Sunday). The use of Powered Mechanical Equipment (PME) for construction works during restricted hours would require a Construction Noise Permit (CNP). The GW-TM details the procedures for assessing such application. The granting of a CNP is subject to the contemporary conditions and it may be revoked at any time for failure to comply with the permit conditions.

4.2.2.3 In addition to the general control on the use of PME during restricted hours, the use of Specified Powered Mechanical Equipment (SPME) and the undertaking of Prescribed Construction Work (PCW) during the restricted hours in a Designated Area are also controlled by the DA-TM. Construction plant or equipment classified as SPME under the DA-TM includes hand-held breaker, bulldozer, concrete lorry mixer, dump truck and hand-held vibratory poker. The PCW includes the erection or dismantling of formwork or scaffolding, hammering, loading, unloading or handling of rubble, wooden boards, steel bars, wood or scaffolding material.

4.2.2.4 The DA-TM details the procedures that are adopted by the Noise Control Authority for assessing the use of SPME during restricted hours and determining whether a CNP could be issued.

4.2.2.5 Noise from construction activities during restricted hours at the Noise Sensitive Receivers (NSRs) are controlled under the TMs and shall not generally be exceeded the specified Acceptable Noise Levels (ANLs). These ANLs are stipulated in accordance with the Area Sensitivity Ratings (ASRs) established for the NSRs. The ASR is determined according to the characteristics of the area within which the NSR is located, and the degree of the effect on the NSR by the "influencing factors" (IF) (such as the presence of industrial area or major roads). The ANLs for construction works in Designated Areas, predominantly populated built-up areas, are more stringent (i.e. 15 dB(A) lower than those listed in the GW-TM). The ANLs for construction works during restricted hours are summarized in **Table 4.2**.

Table 4.2 Acceptable Noise Levels for Construction Works during Restricted Hours

Time Period	Acceptable Noise Levels (ANLs), dB(A)		
	Area Sensitivity Rating A	Area Sensitivity Rating B	Area Sensitivity Rating C
All days during the evening (1900 to 2300 hours), and general holidays (including Sundays) during the daytime and evening (0700 to 2300 hours)	60 (45)	65 (50)	70 (55)
All days during the night-time (2300 to 0700 hours)	45 (30)	50 (35)	55 (40)

Note:
1. Figures in brackets are ANLs for the use of SPME in Designated Areas.

4.2.2.6 As defined in the Designated Areas – Yuen Long, Tin Shui Wai, Mai Po, Shek Kong and Kwu Tung under the NCO (i.e. Plan No. EPD/AN/NT-01), Kam Tin areas such as Sha Po Tsuen, Park Yoho, Crescent Green and Riva; Ngau Tam Mei areas such as Pok Wai and Long Ha; Kwu Tung areas such as Europa Garden and Valais; fall within the Designated Area (DA).

4.2.2.7 The Noise Control Authority will consider a well-justified CNP application, for construction works within restricted hours as guided by the relevant TMs issued under the NCO. The Noise Control Authority will take into account contemporary conditions / situations of adjoining land uses and any previous complaints against construction activities at the site before deciding whether to grant a CNP. Nothing in the EIA Report should bind the Noise Control Authority in making its decision. If a CNP is to be issued, the Noise Control Authority should include in the permit any condition it considers appropriate. Failure to comply with any such conditions will lead to cancellation of the CNP and prosecution under the NCO.

Percussive Piling

4.2.2.8 Percussive piling is prohibited between 1900 and 0700 hours on any weekday not being a general holiday and at any time on Sunday or general holiday under the NCO. A CNP is required for carrying out percussive piling between 0700 and 1900 hours on any day not being a general holiday. PP-TM sets out the requirements for working and

determination of the permitted hours of operations for the CNP applications. The permitted hours of operations would be 3, 5 or 12 hours per day depending on the types of percussive piling and the predicted noise impact at NSRs.

Blasting

4.2.2.9 There is no statutory procedure or criteria under the NCO and EIAO for assessing the blasting impacts and is therefore beyond the scope of the EIA. However, the administrative and procedural control of all blasting operations in Hong Kong is vested in the Mines Division of the Civil Engineering and Development Department (CEDD). The Dangerous Goods Ordinance (Cap. 295) also stipulates that no person shall carry out blasting unless he possesses a valid mine blasting certificate to be issued by the Mines Division of CEDD. The Superintendent of Mines will review the application on a case-by-case basis before issuing the Mine Blasting Certificate. Although there is no statutory noise level for blasting, the noise associated with the removal of debris and rocks are controlled under the EIAO-TM.

Construction Groundborne Noise during Non-restricted Hours

4.2.2.10 Noise arising from general construction works during normal daytime (i.e. 0700 – 1900 hours on any day not being a Sunday or general holiday) is governed by the EIAO-TM. With reference to the IND-TM under the NCO, the criteria for noise transmitted primarily through the structural elements of the building, or buildings, should be 10 dB(A) less than the relevant ANL (i.e. ANL - 10 dB(A)). These criteria apply to all NSRs, such as residential buildings, schools, clinics, hospitals, temples and churches. Therefore, the construction groundborne noise criteria are limited to 10 dB(A) below the respective ANL stipulated in the EIAO-TM. A summary of construction groundborne noise criteria during normal daytime is given in **Table 4.3**.

Table 4.3 Construction Groundborne Noise Criteria for Non-restricted Hours

Uses	Daytime Construction Groundborne Noise Criteria, Leq (30 mins) dB(A)
• All domestic premises; • Temporary housing accommodation; • Hostels; • Convalescent homes; and • Homes for the aged	65
• Places of public worship; • Courts of law; and • Hospitals and medical clinics	60
• Educational institutions (including kindergartens and nurseries)	60 55 (during examinations)

Construction Groundborne Noise during Restricted Hours

4.2.2.11 As discussed in the above section, same principle for groundborne noise criteria (i.e. ANL - 10 dB(A) under the IND-TM) shall be adopted. Therefore, **Table 4.4** below summarizes the construction groundborne noise criteria during restricted hours.

Table 4.4 Construction Groundborne Noise Criteria for Restricted Hours

Time Period	Noise Criteria, dB(A)		
	Area Sensitivity Rating A	Area Sensitivity Rating B	Area Sensitivity Rating C
All days during the evening (1900 to 2300 hours), and general holidays (including Sundays) during the daytime and evening (0700 to 2300 hours)	50	55	60
All days during the night-time (2300 to 0700 hours)	35	40	45

4.2.3 Operation Noise

4.2.3.1 The EIAO–TM (Annex 5 of TM) has stipulated the noise standards for various noise sources. Noise standards relevant to the operation of the Project are shown in the following **Table 4.5**.

Table 4.5 Noise Standards for Operational Phase

Common Uses	Noise Standards [1], [2]	
	Road Traffic Noise L_{10} (1hour), dB(A)	Fixed Noise Sources
• All domestic premises; • Temporary housing accommodation; • Hostels; • Convalescent homes; and • Homes for the aged	70	(a) 5dB(A) below the appropriate ANLs shown in the Technical Memorandum for the Assessment of Noise from Places Other than Domestic Premises, Public Places or Construction Sites, or (b) the prevailing background noise levels (For quiet areas with noise level 5 dB(A) below the ANL)
• Educational institutions (including kindergartens and nurseries); • Places of public worship; and • Courts of law	65	
• Hospitals and medical clinics	55	
Note: [1] The above standards, or equivalent, apply to uses that rely on opened windows for ventilation and are assessed at 1m from the external façade.		

Road Traffic Noise

4.2.3.2 The assessment and designing of mitigation measures against road traffic noise impact shall also follow EIAO GN No. 12/2023 on “Road Traffic Noise Impact Assessment under the Environmental Impact Assessment Ordinance”.

Fixed Noise Sources

4.2.3.3 Operational noise from fixed noise sources is controlled under the IND-TM. For assessing fixed noise sources, the ASRs at the NSRs are defined in accordance with the relevant TMs. To plan for a better environment, the EIAO-TM has specified the following requirements for the planned fixed noise sources, whichever is more stringent.

- 5dB(A) below the appropriate ANLs in the IND-TM; or
- the prevailing background noise levels.

4.2.3.4 The determination of the ASR for the NSR as stipulated in IND-TM is provided in **Table 4.6**.

Table 4.6 ASRs in IND-TM

Degree to which NSR is affected by Influencing Factor (IF) / Type of Area Containing NSR	Not Affected	Indirectly Affected	Directly Affected
(i) Rural area, including country parks or village type developments	A	B	B
(ii) Low density residential area consisting of low-rise or isolated high-rise developments	A	B	C
(iii) Urban area	B	C	C
(iv) Area other than those above	B	B	C

Notes:
[1:] “Country park” means an area that is designated as a country park pursuant to section 14 of the Country Parks Ordinance;
[2]: “Directly affected" means that the NSR is at such a location that noise generated by the IF is readily noticeable at the NSR and is a dominant feature of the noise climate of the NSR;
[3]: "Indirectly affected" means that the NSR is at such a location that noise generated by the IF, whilst noticeable at the NSR, is not a dominant feature of the noise climate of the NSR;
[4]: "Not affected" means that the NSR is at such a location that noise generated by the IF is not noticeable at the NSR; and
[5]: "Urban area" means an area of high density, diverse development including a mixture of such elements as industrial activities, major trade or commercial activities and residential premises.

4.2.3.5 The ANLs for different ASRs during different periods are summarised in the **Table 4.7**.

Table 4.7 ANLs for Fixed Noise Sources

Time Period	ANL, dB(A)		
	Area Sensitivity Rating A	Area Sensitivity Rating B	Area Sensitivity Rating C
Day (0700 to 1900 hours)	60	65	70

Time Period	ANL, dB(A)		
	Area Sensitivity Rating A	Area Sensitivity Rating B	Area Sensitivity Rating C
Evening (1900 to 2300 hours)	60	65	70
Night (2300 to 0700 hours)	50	55	60

4.3 Baseline Conditions

4.3.1 Description of Noise Environment

4.3.1.1 The Project is an approximately 9km long dual three-lane carriage, mainly in the form of at-grade road, tunnel and viaduct connecting between the interchange of San Tin Highway and Tsing Long Highway via Ngau Tam Mei to Kwu Tung.

4.3.1.2 Interchanges will be constructed at Yuen Long Highway/Tsing Long Highway, Ngau Tam Mei Development Area, San Tin Technopole, Fanling Highway and Kwu Tung North NDA. The noise climate at these interchanges are dictated by the road traffic noise from their surrounding road network, i.e. noise climate at the interchange at Yuen Long Highway/Tsing Long Highway is dominated by the road traffic from Yuen Long Highway and Tsing Long Highway; noise climate at Ngau Tam Mei Development Area is dictated by the scattered industrial or brownfield operations and from the road traffic from local roads; noise climate at San Tin Technopole is dictated by the scattered industrial or brownfield operations and from the road traffic from the local road; and the noise climate at Kwu Tung North NDA is dictated by the road traffic from the road network including Fanling Highway.

4.3.1.3 Tunnel will be constructed from Ngau Tam Mei to San Tin Technopole and from San Tin Technopole to Kwu Tung area. The areas which tunnel passed through are conservation area at Ngau Tam Mei and Kwu Tung Reservoir respectively, which are mainly rural in nature. The noise climate at these areas is dictated by scattered industrial or brownfield operations and from road traffic from local roads.

4.3.1.4 Open road sections will be constructed from Tung Shing Lei to Ngau Tam Mei and Kwu Tung North. The areas are comprises of greenbelts, comprehensive development areas, villages, industrial areas and conservation areas. The noise climate at the open road sections between Tung Shing Lei and Ngau Tam Mei is dictated by Yuen Long Highway, Tsing Long Highway, San Tam Road and Castle Peak Road – Tam Mi; and the noise climate along Ngau Tam Mei New Development Area (NTM NDA) and San Tin Technopole is dictated by scattered industrial or brownfield operations and from road traffic from local roads. Noise climate for the section at Kwu Tung North is dictated by Fanling Highway, Castle Peak Road – Chau Tau and Kwu Tung Road.

4.3.2 Prevailing Noise Measurements

4.3.2.1 Prevailing background noise at NTM NDA was measured and reported in the approved Northern Link (NOL) EIA report (Register No.: AEIAR- 259/2024). A site survey was conducted in July 2025 to verify the applicability of these prevailing background noise measurements conducted in 2022/2023. Based on the site survey findings and a review of existing land uses, no significant changes were identified in the environmental context in the vicinity of the NTM NDA. The existing noise climate of the NTM NDA is expected to remain similar to the conditions during the prevailing

background noise measurement presented in the NOL Main Line EIA report, therefore, the measurement results are considered applicable to the Project. The prevailing background noise levels for the NTM NDA were referenced from Appendix 4.1 of the approved NOL Main Line EIA report, and the relevant results are extracted and presented in **Table 4.8**. This prevailing noise measurement result is adopted to identify the fixed noise source impact assessment criteria for conservative assessment.

4.3.2.2 Prevailing noise measurements have also been conducted near Ngau Tam Mei Water Treatment Works and Ki Lun Tsuen in order to capture the existing noise environment near portal plant room near the tunnels. The prevailing noise measurements were conducted in July 2025 and the measurement locations are shown in **Appendix 4.1**. L_{90} was used as the parameter to establish the corresponding noise criteria according to HKPSG. A summary of the results is given in **Table 4.8**. These prevailing noise measurement results are adopted to identify the fixed noise sources impact assessment criteria for conservative assessment. **Appendix 4.1** shows the detailed prevailing noise measurement results.

Table 4.8 Prevailing Noise Level Measurement

Measurement Location & Description	Time Period	Corrected Noise Level, dB(A) [1]
M06 - Ngau Tam Mei [2]	Daytime (0700 – 1900) and Evening (1900 – 2300)	47 - 48
	Night-time (2300 – 0700)	42
NML1 - Ngau Tam Mei	Daytime (0700 – 1900) and Evening (1900 – 2300)	41
	Night-time (2300 – 0700)	38
NML2: Ki Lun Tsuen South	Daytime (0700 – 1900) and Evening (1900 – 2300)	50
	Night-time (2300 – 0700)	48
NML3: Ki Lun Tsuen North	Daytime (0700 – 1900) and Evening (1900 – 2300)	49
	Night-time (2300 – 0700)	48
Notes: [1] Noise level with a 3dB(A) façade correction. [2] Extracted from Northern Link EIA Report conducted in 2022/2023		

4.4 Noise Sensitive Receivers

4.4.1 Identification of Noise Sensitive Receivers

4.4.1.1 With reference to Annex 13 of the EIAO-TM, NSRs include residential uses (all domestic premises, temporary housing accommodation, hostels), educational institutions (including kindergarten and nurseries), hospitals, medical clinics, homes for the aged, convalescent homes, places of worship, courts of law and any others premises or places that are considered by the Director to have similar sensitivity to noise as the above. These NSRs cover the existing sensitive developments, committed and planned NSRs during construction and operation phases of the Project.

4.4.1.2 The assessment area for the Project is defined as 300m away from the Project boundary in accordance with S.3.5.4.2 of the EIA Study Brief; S.1.1.1.(a), S.2.1.1(1) and S.3.1.1(a) in Appendix C of the EIA Study Brief. Representative NSRs within the assessment area have been identified with the most affected layer of NSRs. The openable windows for ventilation of the representative NSRs are selected as Noise Assessment Points (NAPs) for assessment.

4.4.1.3 The existing NSRs are identified by means of topographic maps, aerial photos, land status plans and site inspections in July 2025Planned / committed NSRs within the Assessment Area are also identified by making reference to relevant documents as listed:

- Nam Sang Wai Outline Zoning Plan (OZP) (No. S/YL-NSW/11 dated 12 September 2025);
- Kam Tin North OZP (No. S/YL-KTN/11 dated 15 December 2023);
- Kam Tin South OZP (No. S/YL-KTS/15 dated 21 December 2018);
- Ngau Tam Mei OZP (No. S/YL-NTM/15 dated 31 October 2025);
- Kwu Tung South OZP (No. S/NE-KTS/22 dated 21 February 2025);
- San Tin Technopole OZP (No. S/STT/2 dated 20 September 2024);
- Kwu Tung North OZP (No. S/KTN/4 dated 29 September 2023);
- Planning Applications under S.16 / S.12a Town Planning Ordinance;
- Development Permission Area Plans and Outline Development Plans by the Planning Department (PlanD);
- Land Sale Programme published by the Lands Department (LandsD); and
- Other relevant published plans, including plans, drawings and applications by the correspondent project proponents.

4.4.1.4 The tentative programme of identified planned developments within the 300m assessment area of the Project is summarised in **Table 4.9** below.

Table 4.9 Tentative Programme of Identified Planned Developments

No.	Project	Tentative Programme	
		Commencement	Completion
1	CE8/2022 – Site Formation and Infrastructure Works for Public Housing Development at Sha Po, Shap Pat Heung and Tai Kei Leng Yuen Long	Under review by the Government	
2	Ngau Tam Mei New Development Area under CE33/2021 – Land Use Review Study for Ngau Tam Mei Area Feasibility Study	2027	2036
3	San Tin Technopole under CE 20/2021 – First Phase Development of the New Territories South – San Tin / Lok Ma Chau Development Node – Investigation	2024	2039

No.	Project	Tentative Programme	
		Commencement	Completion
4	Kwu Tung North New Development Area under remaining phase of site formation and engineering infrastructure works at Kwu Tung North New Development Area and Fanling North New Development Area	2013	2031
5	Planned residential development under CE33/2021 – Land Sharing Pilot Scheme Application No. LSPS/002	No publicly available information	2031
6	Planned residential development under Planning Application no. Y/YL-NSW/7	2026	2030
7	Planned residential development under Planning Application no. Y/YL-NSW/8	No publicly available information	2031
8	Planned residential development under Planning Application no. Y/YL-NSW/9	No publicly available information	2031
9	Planned Residential Development under Planning Application no. A/YL-NSW/314	No publicly available information.	
10	Planned residential development under Planning Application no. A/YL-NSW/293	No publicly available information.	
11	Planned residential development under Planning Application no. Y/YL-KTN/964	No publicly available information.	
12	Planned residential development under Planning Application no. A/YL-NSW-348&349	No publicly available information	2030
13	Planned School under the Planning Application no. Y/NE-KTS/18	No publicly available information	2036

4.4.1.5 Assessment area and locations of first layer noise sensitive receivers are shown in **Figure 4.1**. Identified representative NSRs within 300m assessment area for construction noise, road traffic noise and fixed noise sources impacts are listed in **Table 4.10** below. Some other structures within the Project boundary or closer than the identified representative NSRs were inspected. They were of non-noise sensitive use, and thus not considered as representative NSRs. Discussion on the assessment area for construction noise, road traffic noise, and fixed noise sources impact is given in **Section 4.5**, **Section 4.6** and **Section 4.7**, respectively. Photos of the representative existing NSRs are presented in **Appendix 4.2**.

Table 4.10 Representative Noise Sensitive Receivers

NSR Description	Representative NSR ID [1]	NAP	Zoning [2]	Use [3]	No. of Storeys	Construction Noise	Road Traffic Noise	Fixed Noise Sources	Shortest Horizontal Distance from the Project Works Boundary, m
Existing NSRs									
Ko Po San Tsuen	KPST1-3	KPST1, KPST2, KPST3	AGR	R	1-3	✓	✓	-	160
HK Boy Scouts Association Yuen Long East District Centre	HKBSA	HKBSA	GB	G/IC	3	✓	✓	-	280
Au Tau Youth Centre	AT1	AT1	GB	G/IC	1	✓	✓	-	280
Ha Ko Po Tsuen	HKPT	HKPT	R(E)	R	1	✓	✓	-	70
Helorus Boulevard Tower 2 & 3 and	RV1-2	RV1, RV2	R(B)	R	21	✓	✓	-	85
Tower 6, Tower 9 & Tower 10, Crescent Green	CR1-3	CR1, CR2, CR3	R(B)1	R	12	✓	✓	-	22
Village type houses at Tung Shing Lei	TSL1-3	TSL1a, TSL1b, TSL2, TSL3a, TSL3b	U	R	1-2	✓	✓	-	20
Tower 2B, 7A, 1A Park Yoho	PY1-3	PY1a, PY1b, PY2, PY3a, PY3b	CDA	R	10-14	✓	✓	-	100
Village type houses at Fung Kat Heung	FKH	FKHa, FKHb	OU	R	1	✓	✓	-	60
Shum Residence	SR1-3	SR1, SR2, SR3	R(C)3	R, W	2	✓	✓	-	100 – 120
Village type houses at Mo Fan Heung	MFH1-8	MFH1, MFH2, MFH3, MFH4, MFH5, MFH6,	OU, I(D)	R	1-2	✓	✓	-	<10

NSR Description	Representative NSR ID [1]	NAP	Zoning [2]	Use [3]	No. of Storeys	Construction Noise	Road Traffic Noise	Fixed Noise Sources	Shortest Horizontal Distance from the Project Works Boundary, m
		MFH7a, MFH7b, MFH7c, MFH8							
Village type houses at Wing Kei Tsuen	WKT1-2	WKT1, WKT2	OU	R	2	✓	✓	-	110
Elite Garden	EG1-6	EG1, EG2, EG3, EG4, EG5, EG6	V	R	3	✓	✓	-	25
Village type houses at Pok Wai	PW1-6	PW1a, PW1b, PW1c, PW2a, PW2b, PW3a, PW3b, PW4a, PW4b, PW5, PW6	OU	R	3	✓	✓	-	<10
Merry Garden	MG1	MG1	V	R	3	✓	✓	-	20
Village type houses at Long Ha	LH	LHa, LHb	CDA	R	1	✓	✓	-	20
Chan Ming Yam Tong	CMYT1-2	CMYT1a, CMYT1b, CMYT1c, CMYT2	CDA, OU	R	1	✓	✓	-	35
Village type houses at San Wai Tsuen	SWT1-4	SWT1, SWT2a, SWT2b, SWT2c, SWT3a, SWT3b, SWT3c, SWT4	V	R	2-3	✓	✓	-	70
Village type houses at Ngau Tam Mei	NTMV1-4	NTMV1, NTMV2, NTMV3, NTMV4	V	R	1-2	✓	✓	-	15
Village type houses at Cheung Lek	CL1-2	CL1, CL2a, CL2b	GB	R	1	✓	✓	✓	30
Village type houses at Ki Lun Tsuen South	KLTS1-4	KLTS1a, KLTS1b, KLTS2, KLTS3, KLTS4	GB	R	1-2	✓	✓	✓	35

NSR Description	Representative NSR ID [1]	NAP	Zoning [2]	Use [3]	No. of Storeys	Construction Noise	Road Traffic Noise	Fixed Noise Sources	Shortest Horizontal Distance from the Project Works Boundary, m
Village type houses at Ki Lun Tsuen North	KLTN1–4	KLTN1a, KLTN1b, KLTN1c, KLTN2a, KLTN2b, KLTN3a, KLTN3b, KLTN4a, KLTN4b	AGR, GB	R	1-2	✓	✓	✓	20
Village type houses at Kwu Tung	KT1-3	KT1, KT2, KT3	GB	R	1-2	✓	✓	✓	25
Kwu Tung North Multi-welfare Services Complex	KTN1	KTN1	G/IC	R	7	✓	✓	-	60
Europa Garden	EPG1-6	EPG1a, EPG1b, EPG2a, EPG2b, EPG3a, EPG3b, EPG4a, EPG4b, EPG5a, EPG5b, EPG6a, EPG6b	R(C)2	R	2	✓	✓	-	<10
Lady Ho Tung Welfare Centre Eco-Learn Institute	LHTWC	LHTWCa, LHTWCb	G/IC	E	1	✓	✓	-	23
Valais	VL	VL	R(C)2	R	2	✓	✓	-	18
Village type houses at Ma Tso Lung	MCL	MCLa, MCLb, MCLc	GB	R	1	✓	✓	-	160
Planned NSRs									
Proposed Comprehensive Residential Development under	KTN964_1-3	KTN964_1, KTN964_2, KTN964_3	R(E)	R	15	-	✓	-	150

NSR Description	Representative NSR ID [1]	NAP	Zoning [2]	Use [3]	No. of Storeys	Construction Noise	Road Traffic Noise	Fixed Noise Sources	Shortest Horizontal Distance from the Project Works Boundary, m
Application No. A/YL-KTN/964									
Proposed Comprehensive Residential Development under Application No. A/YL-NSW/293	NSW293_1-9	NSW293_1 NSW293_2, NSW293_3, NSW293_4, NSW293_5a, NSW293_5b, NSW293_5c, NSW293_6a, NSW293_6b, NSW293_6c, NSW293_7, NSW293_8, NSW293_9	U	R	3-43	-	✓	-	80
Land Sharing Pilot Scheme (LSPS) Application No. LSPS/002	LSPS1-2	LSPS1a, LSPS1b, LSPS2	R(A)1, R(A)2	R	24	✓	✓	-	210
Planned Public Housing Development and primary school at Sha Po under Agreement No. CE8/2022(CE)	SP1-7	SP1, SP2a, SP2b, SP3, SP4, SP5, SP6, SP7	R(A)	R, E	7-55	-	✓	-	100
Proposed Residential Development and School Site under Application No. Y/YL-NSW/9	NSW9_1-3	NSW9_1a, NSW9_1b, NSW9_2a, NSW9_2b, NSW9_2c, NSW9_2d, NSW9_2e, NSW9_2f, NSW9_3a, NSW9_3b, NSW9_3c, NSW9_3d, NSW9_3e, NSW9_3f	OU	R, E	8-28	✓	✓	-	15 - 50

NSR Description	Representative NSR ID [1]	NAP	Zoning [2]	Use [3]	No. of Storeys	Construction Noise	Road Traffic Noise	Fixed Noise Sources	Shortest Horizontal Distance from the Project Works Boundary, m
Proposed Residential Development under Application No. Y/YL-NSW/8	NSW8_1-3	NSW8_1a, NSW8_1b, NSW8_1c, NSW8_1d, NSW8_1e, NSW8_1f, NSW8_2a, NSW8_2b, NSW8_2c, NSW8_2d, NSW8_3a, NSW8_3b	OU	R	27	✓	✓	-	<10
Proposed Comprehensive Development under Application No. Y/YL-NSW/7	NSW7_1-3	NSW7_1, NSW7_2, NSW7_3,	OU	R	15	✓	✓	-	130
Proposed Social Welfare Facilities (RCHE) under Application No. A/YL-NSW/348&349	NSW348&NSW349	NSW348, NSW349	V	R	3	✓	✓	-	220
Proposed Residential Development under Application No. A/YL-NSW/314	NSW314_1-2	NSW314_1, NSW314_2	OU	R	2	-	✓	-	270
Proposed Residential Development, Staff Quarters and Student Hostels at Ngau Tam Mei New Development Area	NTM1-12	NTM1, NTM2, NTM3, NTM4, NTM5, NTM6, NTM7, NTM8, NTM9, NTM10, NTM11, NTM12	OU, R, G/IC	R	20-47	✓	✓	✓	15

NSR Description	Representative NSR ID [1]	NAP	Zoning [2]	Use [3]	No. of Storeys	Construction Noise	Road Traffic Noise	Fixed Noise Sources	Shortest Horizontal Distance from the Project Works Boundary, m
Proposed School at Kwu Tung South under Application No. Y/NE-KTS/18	KTSIS1-2	KTSIS1, KTSIS2	AGR	R	4-10	✓	✓	-	90
Planned residential developments and schools at Kwu Tung North New Development Area	KTN2-7	KTN2, KTN3, KTN4, KTN5, KTN6, KTN7	G/IC, OU, R(A)1, R(A)2	R, E	8-38	✓	✓	-	30 – 35

Notes:
[1]: The assessment will only include NSRs which rely on opened windows for ventilation, and only the first layer of NSRs has been selected for assessment.
[2]: R: Residential (Private Housing); R(A)1: Residential (Group A)1; R(A)2: Residential (Group A)2; R(B): Residential (Group B); R(B)1: Residential (Group B)1; R(C)2: Residential (Group C)2; R(C)3: Residential (Group C)3; R(E): Residential (Group E); U: Undetermined; CDA: Comprehensive Development Area; G/IC: Government, Institution or Community; OU: Other Specified Uses; I(D): Industrial (Group D); V: Village Type Development; AGR: Agriculture; GB: Green Belt; C: Commercial.
[3]: R: residential; E: educational institution; W: places of public worship, G/IC: Government, Institution and Community
[4]: For the school site Anchors Academy Affiliated International Kindergarten cum Nursery near Yuen Long Highway, an enquiry email has been issued to confirm the means of ventilation. It has been confirmed that the classrooms are equipped with central air conditioning system and do not rely on opened windows for ventilation, corresponding email is provided in **Appendix 4.2**.

4.5 Construction Noise Impact Assessment

4.5.1 Assessment Methodology

4.5.1.1 According to Clause 1.2.2 of Appendix C of the EIA Study Brief, a qualitative assessment is required to demonstrate no adverse construction noise impact would be associated with the Project by adopting quieter construction method and equipment and further construction noise mitigation measures during the construction stages. The general approach of this qualitative construction noise impact assessment, with reference to Sections 5.3 and 5.4 of Annex 13 of the EIAO-TM and EIAO GN No. 9/2023, is summarized below:

- Determine 300m assessment area from the Project Site boundary which includes the associated works required under the Project;
- Identify existing, committed and planned NSRs within the assessment area and assign NAPs on the representative NSRs that would most likely be affected by noise from the construction works;
- Summarize the construction method for the key construction works;
- Collate the construction plant inventory for the key construction works;
- Evaluate the potential construction noise impact on the NSRs qualitatively;
- Evaluate cumulative construction noise impact from concurrent projects qualitatively;
- Examine and recommend all practical mitigation measures, such as adoption of alternative construction methodology, quiet plant, silencer, enclosure, etc., to alleviate any potential noise impacts as much as practicable with reference to Annex 13 of the EIAO-TM and EIAO GN No. 9/2023; and
- Propose to submit a Construction Noise Management Plan (CNMP), which contains quantitative construction noise impact assessment, to the Environmental Protection Department (EPD).

4.5.2 Identification of Construction Noise Impact

Identification of Assessment Area and NSR

4.5.2.1 According to Clause 1.1.1(a) of Appendix C of the EIA Study Brief, the assessment area for construction noise impact assessment shall include areas within 300m from the boundary of the Project and its associated works (see **Figure 4.2**).

4.5.2.2 Representative NSRs and NAPs that are affected by construction noise impact are identified and presented in **Table 4.10** above and depicted in **Figure 4.2**.

Tentative Construction Programme

4.5.2.3 Construction of the Project is tentatively scheduled to commence in Year 2027 and completed by Year 2036. Tentative programme for commissioning of the Northern Metropolis Highway – San Tin Section will be in Year 2036.

Inventory of Noise Sources

4.5.2.4 **Section 2** of this EIA Report has described the key project elements and the associated construction methods. The potential sources of noise impact during the construction of the Project would be the use of PME for various construction activities. The main construction works encompass the following:

- Ground investigation (GI) works;
- Site clearance and formation;
- Construction of at-grade roads, viaducts and interchanges;
- Construction of tunnels by drill and blast / drill and break method;
- Construction of Administration Building, Portal Plant Building/ Plant Room/ Control Room; and
- Ancillary works including slope works, adits and associated connection with existing roads, reprovisioning and reinstatement works, landscaping works, road lighting, etc.

4.5.2.5 The construction programme currently envisioned does not require the use of percussive piling. In accordance with the list of good practices and state-of-the-art technologies within the industry, the construction methods and equipment below can be employed in this Project:

- Quieter type blade saws and hydraulic crushers for concrete removal during site clearance;
- Non-explosive chemical expansion agents and hydraulic splitters for concrete or rock breaking activities;
- Self-compacting concrete instead of vibratory poker for concreting works;
- Quieter piling method by large diameter bored piling to minimize the impact on existing structures; and
- Off-site concreting works for viaduct construction with pre-cast segments.

4.5.2.6 According to the construction methods envisaged at this stage, the key construction activities and the associated tentative PMEs have been outlined in **Table 4.11** below. The plant inventory adopted for the assessment have been confirmed by Project Engineer. It should be noted that the Contractor will review and update this tentative construction plant inventory and the Construction Noise Management Plan (CNMP) based on the available engineering data and considerations at that time.

4.5.2.7 For the construction of tunnel, Tunnel Boring Machine (TBM) is not adopted in view of cost-effectiveness which requires relatively higher capital cost of initial investment and limitation on the road alignment (e.g. requiring large turning radius). There is limitation of site areas for launching and retrieving the TBM. Moreover, tunnel section (circular shape) created by TBM would necessarily generate more C&D material than section (horseshoe) created by drill and blast method.

Table 4.11 Tentative Plant Inventory for Key Construction Activities

Activities	Possible PMEs Required	
GI works	• Crane • Drill rig	• Generator
Site clearance and formation	• Backhoe • Breaker • Bulldozer • Compactor • Crane • Drilling plant • Dump truck	• Excavator • Generator • Hand-held breaker • Hydraulic crusher • Loader • Piling machine • Saw
Construction of temporary noise enclosure	• Crane • Generator	• Hydraulic crusher
Construction of at-grade roads, viaducts and interchanges	• Air compressor • Asphalt paver • Bar bender and cutter • Compactor • Concrete lorry mixer • Concrete pump • Crane • Dump truck • Excavator	• Generator • Launching gantry • Paint liner marker • Piling machine • Roller • Trailer truck • Vibratory poker • Water pump
Construction of tunnels by drill and blast / drill and break method	• Batching plant • Breaker • Concrete lorry mixer • Concrete mixer • Concrete pump • Conveyor belt • Crane • Drilling machine • Drill rig • Dump truck • Dumper	• Excavator • Generator • Grader • Grinder • Hydraulic crawler mounted rock drill • Lorry • Shotcrete machine • Tunnelling jumbo • Ventilation fan
Construction of portal structure	• Air compressor • Asphalt paver • Bar bender and cutter • Compactor • Concrete lorry mixer • Concrete pump • Crane	• Dump truck • Excavator • Generator • Roller • Vibratory poker • Water pump
Construction of Administration Building, Portal Plant Building/ Plant Room	• Air compressor • Bar bender and cutter • Concrete lorry mixer • Concrete pump • Crane • Dump truck • Excavator • Generator	• Grout mixer • Grout pump • Lorry • Piling machine • Vibrating hammer • Vibratory poker • Water pump
Slope works	• Breaker • Concrete lorry mixer • Compactor • Drill • Drill rig	• Grout mixer • Grout pump • Hydraulic crusher • Lorry • Saw

Activities	Possible PMEs Required	
	• Dump truck • Excavator • Generator	• Vibratory poker • Water pump
Reprovisioning and reinstatement works	• Air compressor • Asphalt paver • Bar bender and cutter • Compactor • Concrete lorry mixer • Concrete pump • Crane • Dump truck • Excavator • Generator	• Grout mixer • Grout pump • Paint liner marker • Piling machine • Roller • Trailer truck • Vibrating hammer • Vibratory poker • Water pump
Landscaping works	• Breaker • Crane • Dump truck • Excavator	• Generator • Lorry • Saw

4.5.2.8 Apart from the listed PMEs above, it is anticipated that the construction of tunnels would involve drill and blast operations during restricted hours. The Contractor shall obtain the CNPs before commencing any construction works during restricted hours to ensure compliance and effective noise management.

4.5.3 Evaluation of Construction Noise Impact

Kam Tin Area

4.5.3.1 Key construction work sites in Kam Tin area include those near Ha Ko Po Tsuen, Ko Po San Tsuen, Au Tau, Tung Shing Lei, Nam Sang Wai, Park Yoho, Fung Kat Heung, Mo Fan Heung, Wing Kei Tsuen and over Kam Tin River.

4.5.3.2 For the work sites in Kam Tin area, key construction activities include site formation, construction of at-grade roads and interchanges, and construction of columns and decks for viaduct sections. Existing NSRs in the vicinity include Riva, Crescent Green, Park Yoho, Au Tau Youth Centre, HK Boy Scouts Association Yuen Long East District Centre, Shum Residence, and village type houses at Ko Po San Tsuen, Ha Ko Po Tsuen, Tung Shing Lei, Fung Kat Heung, Mo Fan Heung and Wing Kei Tsuen. All these NSRs are 10 to 23 storeys high except the village type houses are 1 to 3 storeys high. The shortest horizontal distances from building façade to the Project works boundary are ranging from approximately <10m to 280m. Based on the PMEs anticipated at this stage (see **Table 4.11**), with the implementation of good site practices including locating mobile plant as far away from NSRs as possible and practicable, the use of Quality Powered Mechanical Equipment (QPME) / quieter construction methods, and when required, additional mitigation measures, such as noise barriers, noise enclosures, etc., no adverse construction noise impacts is anticipated for most NSRs.

4.5.3.3 However, the NSRs at Tung Shing Lei and Mo Fan Heung are located relatively close to the Project works boundary (i.e. approximately <10m to 20m), there would be a need of purposely built noise barrier or site hoarding with higher surface density of 10 kg/m^2 and height of minimum 2.4m, and noise enclosures for relatively stationary PME, to screen off the works fronts and provide extra noise attenuation to protect these NSRs.

4.5.3.4 For the construction of viaduct sections and road works on the viaduct sections, once the slab of the viaduct sections is constructed, the viaduct slab will provide noise screening to the NSRs at Ha Ko Po Tsuen, Ko Po San Tsuen, Au Tau Youth Centre, HK Boy Scouts Association Yuen Long East District Centre, Tung Shing Lei, Fung Kat Heung, Shum Residence, Mo Fan Heung and Wing Kei Tsuen for the construction activities above the viaduct slab. Hence, adverse construction noise impacts to the village type houses and institutions in Kam Tin area during construction at viaduct deck level is not anticipated. Besides, the construction method has duly considered the opportunities of optimizing the site works as much as practicable, and hence the use of pre-cast bridge decking would be adopted as much as practicable, that would further reduce the construction noise impacts.

4.5.3.5 Planned residential developments under Land Sharing Pilot Scheme Application No. LSPS/002, Planning Application no. Y/YL-NSW/8 and Y/YL-NSW/9 will be completed by Year 2031, and thus the NSRs at these sites will be subject to the noise generated from the construction of at-grade roads, interchanges and viaduct sections upon population intake. The shortest horizontal distances from the building façade to the Project works boundary are ranging from approximately <10m to 210m. Based on the PMEs anticipated at this stage (see **Table 4.11**), with the implementation of good site practices including locating mobile plant as far away from NSRs as possible and practicable, the use of Quality Powered Mechanical Equipment (QPME) / quieter construction methods, and when required, additional mitigation measures, such as noise barriers, noise enclosures, etc., no adverse construction noise impacts is anticipated for the NSRs located at the planned residential developments under Land Sharing Pilot Scheme Application No. LSPS/002.

4.5.3.6 Nevertheless, the NSRs at the planned residential developments under Planning Application no. Y/YL-NSW/8 and Y/YL-NSW/9 would be relatively closer to the Project works boundary (i.e. approximately <10m to 50m), there would be a need of purposely built noise barriers or site hoarding with higher surface density of 10 kg/m^2 and height of minimum 2.4m, and noise enclosures for relatively stationary PME, to screen off the works fronts and provide extra noise attenuation to protect these NSRs.

4.5.3.7 Construction vehicles would also be required to transport various materials to and from the construction sites as necessary. In order to minimize the loading on existing roads (e.g. Kam Tin Road and Castle Peak Road), the Contractor will implement measures to ensure construction traffic would avoid the peak hours where practicable. This would largely help to minimize nuisance from both traffic and environmental perspectives. It is also noted that the traffic induced by various construction works in Kam Tin area would vary as the construction progresses. For example, the construction traffic generated during the initial and later stages would likely be less than those during the peak construction period which would constitute a relatively shorter period. The Detailed Designer and the Contractor shall review all the contemporary issues (e.g. constructability, site constraints, detailed GI information, etc.) to optimize the construction methods and the generation of construction vehicles.

Ngau Tam Mei Area

4.5.3.8 Key construction work sites in Ngau Tam Mei area include those near Pok Wai, Long Ha, San Wai Tsuen, Chuk Yau Road and Ngau Tam Mei Water Treatment Works.

4.5.3.9 For the work sites in Ngau Tam Mei area, key construction activities include site formation, construction of at-grade roads and interchanges, construction of columns and decks for viaduct sections, construction of portal structures and tunnel sections,

construction of Administration Building, and slope works. Existing NSRs in the vicinity include Elite Garden, Merry Garden, Chan Ming Yam Tong and village type houses at Pok Wai, Long Ha, San Wai Tsuen and Ngau Tam Mei. All these NSRs are 1 to 3 storeys high. The shortest horizontal distances from building façade to the Project works boundary are ranging from approximately <10m to 70m. Based on the PMEs anticipated at this stage (see **Table 4.11**), with the implementation of good site practices including locating mobile plant as far away from NSRs as possible and practicable, the use of QPME / quieter construction methods, and when required, additional mitigation measures, such as noise barriers, noise enclosures, etc., no adverse construction noise impacts is anticipated.

4.5.3.10 However, the NSRs at Pok Waiand Long Ha would be relatively closer to the Project works boundary (i.e. approximately <10m to 20m), and for the slope works in the vicinity of Pok Wai, Long Ha, Chan Ming Yan Tong, San Wai Tsuen and Ngau Tam Mei Development Area, there would be a need of purposely built noise barrier or site hoarding with higher surface density of 10 kg/m^2 and height of minimum 2.4m, and noise enclosures for relatively stationary PME, to screen off the works fronts and provide extra noise attenuation to protect these NSRs.

4.5.3.11 For the construction of viaduct sections and road works on the viaduct sections, once the slab of the viaduct sections is constructed, the viaduct slab will provide noise screening to the village type houses at Pok Wai, Elite Garden, Merry Garden, Long Ha, Chan Ming Yam Tong, San Wai Tsuen and Ngau Tam Mei for the construction activities above the viaduct slab. Hence, adverse construction noise impacts to these NSRs during construction at viaduct deck level is not anticipated. Besides, the construction method has duly considered the opportunities of optimizing the site works as much as practicable, and hence the use of pre-cast bridge decking would be adopted as much as practicable, that would further reduce the construction noise impacts.

4.5.3.12 Planned residential developments under Planning Application no. Y/YL-NSW/7 and A/YL-NSW-348&349 will be completed by Year 2030, and thus the NSRs at these sites will be subject to the noise generated from the slope works and construction of at-grade roads and viaduct sections upon population intake. The shortest horizontal distances from the building façade to the Project works boundary are ranging from approximately 130m to 220m. Based on the PMEs anticipated at this stage (see **Table 4.11**), with the implementation of good site practices including locating mobile plant as far away from NSRs as possible and practicable, the use of Quality Powered Mechanical Equipment (QPME) / quieter construction methods, and when required, additional mitigation measures, such as noise barriers, noise enclosures, etc., no adverse construction noise impacts is anticipated.

4.5.3.13 According to the tentative development programme of Ngau Tam Mei NDA, the first population intake is scheduled in Year 2033. Hence, the NSRs at Ngau Tam Mei NDA will be subject to the noise generated from the slope works and construction of viaduct sections, portal structures and tunnel sections upon population intake. The shortest horizontal distance from the building façade to the Project works boundary is approximately 15m. Based on the PMEs anticipated at this stage (see **Table 4.11**), it is considered that mitigation measures of purposely built noise barriers or site hoarding with higher surface density of 10 kg/m^2 and height of minimum 2.4m, noise enclosures for relatively stationary PME, together with good site practices including locating mobile plant as far away from NSRs as possible and practicable, the use of QPME / quieter construction methods, etc., would be required to minimize the associated construction

noise impacts.

4.5.3.14 Construction vehicles would also be required to transport various materials to and from the construction sites as necessary. In order to minimize the loading on existing roads (e.g. Castle Peak Road and San Tam Road), the Contractor will implement measures to ensure construction traffic would avoid the peak hours where practicable. This would largely help to minimize nuisance from both traffic and environmental perspectives. It is also noted that the traffic induced by various construction works in Ngau Tam Mei area would vary as the construction progresses. For example, the construction traffic generated during the initial and later stages would likely be less than those during the peak construction period which would constitute a relatively shorter period. The Detailed Designer and the Contractor shall review all the contemporary issues (e.g. constructability, site constraints, detailed GI information, etc.) to optimize the construction methods and the generation of construction vehicles.

San Tin Area

4.5.3.15 Key construction work sites in San Tin area include those near Cheung Lek and Ki Lun Tsuen North and South.

4.5.3.16 For the work sites in San Tin area, key construction activities include site formation (including slope works), construction of interchanges, construction of columns and decks for viaduct sections, construction of portal structures and tunnel sections, construction of portal plant buildings and control room, and slope works. Existing NSRs in the vicinity include village type houses at Cheung Lek and Ki Lun Tsuen North and South. All these NSRs are 1 to 2 storeys high. The shortest horizontal distances from building façade to the Project works boundary are ranging from approximately 20m to 35m. Based on the PMEs anticipated at this stage (see **Table 4.11**), it is considered that mitigation measures of purposely built noise barriers or site hoarding with higher surface density of 10 kg/m^2 and height of minimum 2.4m together with good site practices including locating mobile plant as far away from NSRs as possible and practicable, the use of QPME / quieter construction methods, etc. would be required to minimize the associated construction noise impacts.

4.5.3.17 For the construction of viaduct sections and road works on the viaduct sections, once the slab of the viaduct sections is constructed, the viaduct slab will provide noise screening to the NSRs at Cheung Lek and Ki Lun Tsuen North and South for the construction activities above the viaduct slab. Hence, adverse construction noise impacts to these NSRs during construction at viaduct deck level is not anticipated. Besides, the construction method has duly considered the opportunities of optimizing the site works as much as practicable, and hence the use of pre-cast bridge decking would be adopted as much as practicable, that would further reduce the construction noise impacts.

4.5.3.18 Proposed school under Planning Application no. Y/NE-KTS/18 will be completed by Year 2036, and thus the NSRs at this site may be subject to the noise generated from the slope works and road works on interchanges and viaduct sections upon population intake. The shortest horizontal distance from the building façade to the Project works boundary is approximately 90m. Based on the PMEs anticipated at this stage (see **Table 4.11**), with the implementation of good site practices including locating mobile plant as far away from NSRs as possible and practicable, the use of Quality Powered Mechanical Equipment (QPME) / quieter construction methods, and when required, additional mitigation measures, such as noise barriers, noise enclosures, etc., no adverse construction noise impacts is anticipated.

4.5.3.19 Construction vehicles would also be required to transport various materials to and from the construction sites as necessary. In order to minimize the loading on existing roads (e.g. Kwu Tung Road and Fanling Highway), the Contractor will implement measures to ensure construction traffic would avoid the peak hours where practicable. This would largely help to minimize nuisance from both traffic and environmental perspectives. It is also noted that the traffic induced by various construction works in San Tin area would vary as the construction progresses. For example, the construction traffic generated during the initial and later stages would likely be less than those during the peak construction period which would constitute a relatively shorter period. The Detailed Designer and the Contractor shall review all the contemporary issues (e.g. constructability, site constraints, detailed GI information, etc.) to optimize the construction methods and the generation of construction vehicles.

Kwu Tung Area

4.5.3.20 Key construction work sites in Kwu Tung area include those near Kwu Tung Reservoir, Europa Garden, Kwu Tung North Multi-welfare Services Complex, Lady Ho Tung Welfare Centre Eco-Learn Institute, Valais, planned residential developments and schools at Kwu Tung North NDA and Ma Tso Lung.

4.5.3.21 For the work sites in Kwu Tung area, key construction activities include site formation, construction of at-grade roads and interchanges, construction of columns and decks for viaduct sections, construction of portal structures and tunnel sections, and slope works. NSRs in the vicinity include Europa Garden, Valais, Lady Ho Tung Welfare Centre Eco-Learn Institute, Kwu Tung North Multi-welfare Services Complex and village type houses at Kwu Tung and Ma Tso Lung. All these NSRs are 1 to 2 storeys high except the Kwu Tung North Multi-welfare Services Complex and planned residential developments and schools at Kwu Tung North NDA are 7 to 38 storeys high. The shortest horizontal distances from building façade to the Project works boundary are ranging from approximately <10m to 160m. Based on the PMEs anticipated at this stage (see **Table 4.11**) and considered that part of the works at Kwu Tung area will be screened by natural terrain, with the implementation of good site practices including locating mobile plant as far away from NSRs as possible and practicable, the use of QPME / quieter construction methods, and when required, additional mitigation measures, such as noise barriers, noise enclosures, etc., no adverse construction noise impacts is anticipated.

4.5.3.22 However, as the NSRs at Europa Garden, Valais, Lady Ho Tung Welfare Centre Eco-Learn Institute and village type houses at Kwu Tung would be relatively closer to the Project works boundary (i.e. approximately <10m to 25m) and the nearby Project roads are at similar elevation as the NSRs, there would be a need of purposely built noise barriers or site hoarding with higher surface density of 10 kg/m^2 and height of minimum 2.4m and noise enclosures for relatively stationary PME to screen off the works fronts and provide extra noise attenuation to protect these NSRs.

4.5.3.23 For the construction of viaduct sections and road works on the viaduct sections, once the slab of the viaduct sections is constructed, the viaduct slab will provide noise screening to the NSRs at village type houses at Kwu Tung and Ma Tso Lung for the construction activities above the viaduct slab. Hence, adverse construction noise impacts to these NSRs during construction at viaduct deck level is not anticipated. Besides, the construction method has duly considered the opportunities of optimizing the site works as much as practicable, and hence the use of pre-cast bridge decking would be adopted as much as practicable, that would further reduce the construction

noise impacts.

4.5.3.24 Construction of portal structures and tunnel sections will be carried out near the village type houses at Kwu Tung. In view of the screening provided by natural terrain and the tunnel portal construction will be carried out above the viaduct slab, adverse construction noise impact to these NSRs during construction of portal structures and tunnel sections is not anticipated.

4.5.3.25 Construction vehicles would also be required to transport various materials to and from the construction sites as necessary. In order to minimize the loading on existing roads (e.g. Kwu Tung Road and Fanling Highway), the Contractor will implement measures to ensure construction traffic would avoid the peak hours where practicable. This would largely help to minimize nuisance from both traffic and environmental perspectives. It is also noted that the traffic induced by various construction works in Kwu Tung area would vary as the construction progresses. For example, the construction traffic generated during the initial and later stages would likely be less than those during the peak construction period which would constitute a relatively shorter period. The Detailed Designer and the Contractor shall review all the contemporary issues (e.g. constructability, site constraints, detailed GI information, etc.) to optimize the construction methods and the generation of construction vehicles.

Drill and Blast Operations for Tunnel Construction

4.5.3.26 The initial sections of the tunnels will be excavated by mechanical methods. Drill-and-blast excavation will then be adopted for the construction of remaining tunnel sections starting from approximately 50m beyond the tunnel portal. The drill-and-blast works would possibly be conducted throughout a 24-hour cycle on a daily basis as the worst-case scenario subject to the granting of CNP. For safety reason, a metal blast door would be installed at the portal and this blast door would be closed during the blasting works. According to the latest construction method, blasting would be conducted daily at each workfront which will be subject to the agreement from the Mines Division of CEDD. Blasting should be carried out outside sensitive hours as far as practicable. Considering that the tunnel sections are located at rural area without NSR atop or next to them, the drilling and blasting works will be carried out inside the metal blast door and the blasting works will only last for few seconds per blast and be infrequent, it will not cause adverse airborne nor groundborne construction noise impacts to NSRs in the vicinity.

4.5.3.27 Besides, the Contractor will be required to post highly visible warning notices/signs at suitable locations to warn the public that blasting will take place. These warning signs will be posted near the intended blasting location, even though all blasts will be conducted underground. The Contractor is required to state the blasting date and time on the notice.

Consideration of Cumulative Impacts

4.5.3.28 **Section 2** has identified a list of concurrent projects that would be considered in this EIA to address the cumulative impacts.

4.5.3.29 Careful scheduling of construction activities and grouping of PMEs within a construction site and liaison between contractors of nearby construction sites could avoid multiple noisy construction activities/construction equipment to be conducted/used at the same time within close proximity. Hence, the cumulative construction noise level could be reduced.

4.5.3.30 The Project Proponent will further review the cumulative noise impact caused by the Project and concurrent projects within 300m assessment area in the subsequent CNMP submission at later construction phase to ensure that there is no adverse construction noise impact.

4.5.4 Mitigation of Construction Noise Impact

4.5.4.1 Due to the short separation from NSRs, for some construction areas, without proper noise mitigation measures, construction noise exceedances are anticipated. The following mitigation measures have been considered and confirmed the practicality by the Project Engineer:

- Good site practices to limit noise emissions at the source;
- Use of QPME and quieter construction methods;
- Use of temporary noise barriers, noise enclosure, noise insulating fabric, silencer, etc. to screen noise from construction equipment;
- Install acoustic tunnel door or enclosure at the tunnel portals for construction activities to be carried out inside tunnel; and
- Alternative use of plant items within one work site, wherever practicable.

4.5.4.2 The above mitigation measures would need to be implemented in work sites as good practices where appropriate. Detailed descriptions of these mitigation measures are given in the following sections.

4.5.4.3 For NSRs which are close to the construction sites, it is recommended to install purposely built noise barriers or site hoarding with higher surface density of 10 kg/m^2 and height of minimum 2.4m to provide extra noise attenuation to protect these NSRs.

Good Site Management Practices

4.5.4.4 Good site practice and noise management techniques could considerably reduce the noise impact from construction activities on nearby NSRs. The following measures should be practised during each phase of construction:

- Only well-maintained plant should be operated on-site and plant should be serviced regularly during the construction programme;
- Machines and plants (such as trucks, cranes, etc.) that may be in intermittent use should be shut down between work periods or should be throttled down to a minimum;
- Plants known to emit noise strongly in one direction, where possible, be orientated so that the noise is directed away from nearby NSRs;
- Silencers or mufflers on construction equipment should be properly fitted and maintained during the construction works;
- Mobile plants should be located as far away from NSRs as possible and practicable; and
- Material stockpiles, site office and other structures should be effectively utilised, where practicable, to screen noise from on-site construction activities.

Use of QPME and Quieter Construction Methods

4.5.4.5 Quieter construction methods and equipment would be adopted whenever practicable, including the use of QPME for demolition (e.g. hydraulic crusher), quieter method for

rock/concrete breaking (e.g. quieter type saw, non-explosive chemical expansion agent and hydraulic splitter), non-percussive construction method, quieter piling method by large diameter bored piling, pre-cast segments construction, self-compacting concrete, etc.

4.5.4.6 Review on the application of typical quieter construction methods have been conducted and summarized in **Table 4.12** below. With due consideration of the EPD's list of good practices, the followings are examples of quieter construction methods / equipment that are found applicable to this Project.

- Quieter type saw and hydraulic crusher can be adopted for site clearance and site formation works;
- Large diameter bored piling for the piling works; and
- Use of self-compacting concrete or rubber head poker vibrator instead of conventional vibratory poker.

Table 4.12 Typical Quieter Construction Methods/ Equipment

Construction Activities	Conventional Practices	Alternative Quieter Construction Method/ Equipment	Applicable to the Project (Y/N)
Site clearance and site formation works	Drill and break method using excavator-mounter breaker	Use of hydraulic splitter, hydraulic crusher/ quieter type saw (e.g. diamond wire saw, noise reducing diamond blade saw)	Y
Slope Works	Air compressor, generator and mobile crane	Use of QPME	Y
Piling and substructure works including road works	Percussive piling for piling works; Vibratory poker for road works	Use of large diameter bored piling; Self-compacting concrete or rubber head poker vibrator	Y
Superstructure works	Vibratory poker	Self-compacting concrete or rubber head poker vibrator	Y
Tunneling works	Drill and break method using excavator-mounter breaker	Use of chemical agent (non-explosive blasting)	Y
	Drill and blast method	Use of Tunnel Boring Machine (TBM)	N – requires relatively higher capital cost of initial investment and limitation on the road alignment and generate more C&D material

4.5.4.7 More examples of quieter construction methods/equipment can be referenced to EPD's website (https://www.epd.gov.hk/epd/misc/construction_noise/contents/index.php/en/home2/quieter-construction-methods.html).

4.5.4.8 The Contractor may be able to obtain particular models of plant that are quieter than the PMEs given in the GW-TM. The use of quality PME associated with the construction works is prescribed in EPD's QPME database, which contains the SWLs for quality/quiet PME of various types, brands and models.

4.5.4.9 The use of quieter construction methods and equipment, if necessary, will be further reviewed in the detailed design and construction stages, and incorporated in the CNMP.

Use of Temporary Noise Barriers and Noise Enclosures

4.5.4.10 Movable noise barriers or noise enclosures would be erected to provide screening from the construction plant. Noise barriers would be most effective when located immediately adjacent to the PME and can reduce the noise level by 5 dB(A) for movable PME and by 10 dB(A) for stationary PME. The Contractor shall be responsible for design of the noise barrier with due consideration given to the size of the PME and the requirement of intercepting the line of sight between the NSRs and PME. The barrier material of movable noise barrier should have at least 10 kg/m^2 surface density. Purpose-built acoustics barrier can be used to screen noise from particular items of PME or noisy construction activities. The direct line of sight between the PME and the NSRs should be totally screened by a substantial barrier such that the PME will not be visible when viewed from any window, door or other opening in any facade of the NSR. A noise reduction of 10 dB(A) is assumed for purpose-built noise barrier. Full enclosures can be considered as an alternative to shelter relatively static plant including generator. The provision of full enclosure is expected to provide 15 dB(A) noise reduction for static PME. To ensure the above-mentioned noise reduction performance of the noise barrier, there should be no gap or opening for the noise barrier/enclosure being constructed by acoustic mats.

Install Acoustic Tunnel Door or Enclosure at the Tunnel Portals

4.5.4.11 During the initial stage of construction of tunnel portals, construction noise would be generated from the drill & break method with the use of rock drill, the Contractor shall use quieter construction method and/ or quieter plant and shall be responsible for design of the noise barrier/ noise enclosure with due consideration given to the size of the rock drill and the requirement of intercepting the line of sight between the NSRs and PME. The barrier material of noise barrier/ noise enclosure should have at least 10 kg/m^2 surface density. The number of crawler mounted rock drill in each drilling face shall also be minimized as far as possible and practicable to minimize the construction noise impact before the installation of tunnel blasting door.

4.5.4.12 It is considered that installation of acoustic tunnel door or enclosure at the tunnel portals is an effective mitigation measure for construction works to be conducted inside the tunnels. The acoustic tunnel door or enclosure will be installed at the tunnel portal while the exact location would be subject to site condition. It should be made of acoustic panels and the ventilation openings of the tunnel door or enclosure should also be fitted with silencers. The Contractor should select a proper type of acoustic panel and silencer which can provide necessary noise reduction performance to achieve the full compliance of ANLs.

Alternative Use of Plant Items within one Work Site

4.5.4.13 In practice, some plant items will operate sequentially within the same work site, and certain reduction of the predicted noise impacts could be achieved. However, any additional control on the sequencing of plant will impose a restrictive constraint to the Contractor on the operation and planning of plant items, and the implementation of the requirement would be difficult to be monitored. Hence, sequencing operation of PME has not been taken into consideration.

4.5.5 Residual Impact

4.5.5.1 With the implementation of the proposed mitigation measures, construction noise impact to the NSRs could be effectively controlled and minimized. No residual impact is anticipated.

4.5.6 Construction Noise Management Plan

4.5.6.1 According to Clause 1.4 of Appendix C of the EIA Study Brief, a CNMP shall be prepared and submitted to the Director. The CNMP shall contain the quantitative construction noise impact assessment, the adopted quieter construction methods and equipment, noise mitigation measures and the construction noise impact monitoring and audit programme. It shall also make reference to the updated and identified noise mitigation measures once available. The CNMP shall be submitted no later than 2 months before tender invitation, and in any case before the commencement of construction of the Project, in accordance with the requirements stipulated in Section 6.2 of EIAO GN No. 9/2023.Any technical constraint that would hinder the use of these quieter construction method and equipment shall be evaluated and clearly recorded in the assessment.

4.5.6.2 The CNMP shall include construction works programme, construction methodology, PME list, construction noise impact assessment and list of noise mitigation measures to be adopted. It shall incorporate an implementation schedule clearly listing out the mitigation measures, the implementation party, location and timing of implementation. Mitigation measures recommended and requirements specified in the CNMP shall be fully implemented by the Contactor.

4.5.6.3 Cumulative construction noise impact from the concurrent projects should be considered and evaluated in the CNMP. The status of potential planned developments should be reviewed at the time of preparing the CNMP.

4.6 Road Traffic Noise Impact Assessment

4.6.1 Identification of Road Traffic Noise Impact

Identification of Assessment Area and NSR

4.6.1.1 According to Clause 2.1.1(a) of Appendix C of the EIA Study Brief, the assessment area for road traffic noise impact assessment shall include areas within 300m from the boundary of the Project and its associated works (see **Figure 4.3**).

4.6.1.2 Representative NSRs and NAPs that are affected by road traffic noise impact are identified and presented in **Table 4.10** above and depicted in **Figure 4.3**.

Identification of Project Road

4.6.1.3 Road traffic noise generated from vehicular traffic from the Project Roads, existing roads and other planned/committed roads within the 300m assessment area will be assessed.

4.6.1.4 The road category includes the following items and shown in **Appendix 4.3** and summarized in below **Table 4.13**:

Table 4.13 Road Category

Category	Road
Project Road - New Road	Northern Metropolis Highway – San Tin Section, Kwu Tung Section and Tin Shui Wai Section including main trunk road and associated slip roads
Project Road - Altered Existing Road	Yuen Long Highway, Tsing Long Highway, San Tin Highway, Kwu Tung Road, Fanling Highway, Castle Peak Road – Chau Tau
Other Planned/ Committed Road	Road network at Ngau Tam Mei Development Area, Road network for Sha Po Housing Development, Road network at San Tin / Lok Ma Chau Development Node, Road network at Kwu Tung New Development Area
Existing Road	Tsing Long Highway, Yuen Long Highway, San Tam Road, San Tin Highway, Fanling Highway, Kwu Tung Road, Castle Peak Road – Tam Mi and Castle Peak Road – Chau Tau etc.

4.6.2 Assessment Methodology

4.6.2.1 Road traffic noise impact assessment is predicted based on the method of UK Department of Transport "Calculation of Road Traffic Noise (CRTN)". The predicted noise levels at the sensitive receivers include 2.5dB(A) facade reflection and correction factors of effects due to gradient, distance, view angle, road surface and barriers. The road traffic noise will be presented in terms of noise levels exceeded for 10% of the one-hour period during peak traffic flow (i.e. L10(1hr) dB(A)).

4.6.2.2 Calculations of road traffic noise are based on the peak hourly flow for the maximum traffic projected within a 15 years period upon full operation of the roadworks.

4.6.2.3 As discussed in **Section 2**, the commencement year of the Project would be in Year 2036. The assessment year with maximum traffic projections (morning peak hour traffic flows and vehicle compositions, which is generally higher traffic flows than afternoon peak) within 15 years upon operation of the Project would be Year 2051. Hence, Year 2051 was adopted as assessment year in road traffic noise assessment due to its peak traffic prediction in Traffic Impact Assessment of the Project. The traffic forecast for the prevailing year (i.e. the year before the commencement of construction works which is 2026 without project) and the assessment year is presented in **Appendix 4.4**. The traffic flow forecast and endorsement on traffic data from Transport Department ("TD")

was obtained are enclosed in **Appendix 4.4**. Written confirmation from the traffic consultant is shown in **Appendix 4.4** that TD's agreed methodology and the transport model based on the planning assumptions have been strictly adopted in preparing the traffic forecast data for the road traffic noise impact assessment. Both AM and PM traffic forecast have been assessed in the road traffic noise assessment.

4.6.2.4 According to clause 4.7.6 of EIAO GN No. 12/2023, direct mitigation measures would be proposed for Project roads if adverse noise impact is predicted. If the NSRs are also affected by noise from existing roads and other planned/committed roads, direct mitigation measures are required to reduce the noise from Project Roads to a level that it:

- is not higher than the standard; and
- has no significant contribution (less than 1.0 dB(A)) to the cumulative noise level, if the overall noise level (i.e. noise from the Project Roads together with other existing, planned/committed roads) exceeds the noise standard.

4.6.2.5 If the noise criteria will still be exceeded at the existing NSRs after exhausting direct mitigation measures, provision of acoustic insulation in form of windows and provision of air-conditioning for ventilation should be considered under the ExCo directive "Equitable Redress for Persons Exposed to Increased Noise Resulting from the use of New Roads". Eligibility for indirect technical remedies should be tested against the following three criteria:

(a) The predicted overall noise level, L10(1 hour), from the "Project Roads", together with other traffic noise in the vicinity must be above 70 dB(A) for domestic premises and home for the aged, 65 dB(A) for educational institutions, places of public worship and courts of law, and 55 dB(A) for hospitals and medical clinics.

(b) The predicted overall noise level is at least 1.0 dB(A) more than the prevailing traffic noise level.

(c) The contribution to the increase in the predicted overall noise level from the "Project Road" must be at least 1.0 dB(A).

Existing and Planned Noise Mitigation Measures

4.6.2.6 The characteristics of the road network such as road width, surface type and the use of low noise road surfacing (LNRS), the existing and committed noise mitigation measures including the following have been and reviewed based on desktop study and considered in the assessment as presented in **Appendix 4.5**.

- Traffic Noise Mitigation Plan under Development of Kwu Tung North New Development Area, Remaining Phase – Design & Construction (March 2025)
- Noise mitigation measures proposed under CE 8/2022 (CE) Site Formation and Infrastructure Works for Public Housing Developments at Sha Po, Shap Pat Heung and Tai Kei Leng, Yuen Long – Investigation, Design and Construction (March 2025)
- Noise barriers along Yuen Long Highway, Tsing Long Highway and San Tin Highway
- EIA Report of Development at Ngau Tam Mei Area (AEIAR-271/2025)
- EIA Report of San Tin/Lok Ma Chau Development Node (AEIAR-261/2024)

4.6.2.7 In accordance with HyD Guidance Notes on Road Surface Requirements for Expressways and High Speed Roads (RD/GN/032A), Highly Modified Friction Course (HMFC) is proposed as the standard surfacing material on the high speed road sections of new road projects with design speed of 80km/hr or above and expressway. From the RD/GN/032A, the noise reduction performance of HMFC is comparable to Polymer Modified Friction Course (i.e. PMFC), hence, both HMFC and PMFC can serve as LNRS, which provide a noise reduction performance of approximately 2.5dB(A) compare to concrete paving. Locations of existing and committed LNRS (including HMFC to be implemented at proposed project roads) are presented in **Appendix 4.5**.

4.6.2.8 The standard 0.8m solid parapet along main road and slip road of NM Highway has also been included in the assessment.

Assumptions in At-receiver Noise Mitigation Measures of Existing Residential Developments

4.6.2.9 Based on the site visit and desktop review, the at-receiver noise mitigation measures have been implemented at some of the representative NSRs near Yuen Long Highway, Tsing Long Highway and San Tin Highway, including Riva (NSR ID: RV1 and RV2) , Crescent Green (NSR ID: CR1-3) and Park Yoho (NSR ID: PY1-3), and Valais. The locations of the NAPs are shown in **Figure 4.3**.

4.6.2.10 The residential development Riva, including its selected NAPs at Helorus Boulevard Tower 2 and 3, is designed with a single-aspect layout. The habitable rooms designated for sensitive uses (e.g. bedrooms, living rooms) are positioned away from Yuen Long Highway and Tsing Long Highway. The windows at the façade facing these roads are fitted with fixed glazing or are the doors at the non-habitable room for entering the utilities platform / air-conditioner platform.

4.6.2.11 The residential development Crescent Green, including its selected NAPs at Tower 6, Tower 9 and Tower 10, is designed with a single-aspect layout that habitable rooms designated for sensitive uses (e.g. bedrooms, living rooms) are positioned away from Yuen Long Highway and Tsing Long Highway. Windows on the façade facing these roads are fitted with specially provided glazing, i.e. fixed glazing with only maintenance window for cleaning and maintenance purpose.

4.6.2.12 The residential development Park Yoho, including the selected NAPs at Tower 1A and 1B are equipped with solid balconies with noise reduction of 2 dB(A), fixed glazing with only maintenance window for cleaning and maintenance purpose and architectural fins; whereas fixed glazing with only maintenance window for cleaning and maintenance purpose and architectural fins are provided for the NAPs at Tower 7A.

4.6.2.13 Noise barriers were also provided along the site boundary of Valais facing Fanling Highway and Kwu Tung Road.

Assumptions in At-receiver Noise Mitigation Measures of Planned Developments

4.6.2.14 The locations of representative existing and planned NSRs are shown in **Figure 4.3**. There are several approved/ potential planning applications and planned residential developments which are located in the vicinity of Yuen Long Highway (NSR ID.: KTN964, NSW293, Land Sharing Pilot Scheme (LSPS)), San Tin Highway (NSR ID.: SP1-7, NSW7, NSW8, NSW9, NSW314, NSW348 and NSW349) and Hang Tau (NSR ID.: KTSIS1-2). In their respective planning studies, it is anticipated exceedance of

road traffic noise would be occurred in some planned developments, and certain at-receiver noise mitigation measures, e.g. acoustic window and/ or acoustic balcony, would be applied.

4.6.2.15 According to the EA report for Section 16 application A/YL-NSW/293, acoustic fins, enhanced acoustic windows (baffle type), and enhanced acoustic balconies (baffle type) will be adopted for blocks with predicted noise exceedance. These measures provide an adjusted noise reduction of 6.3 dB(A) and 7.0 dB(A) for bedroom and living/dining room respectively after considered the room size correction.

4.6.2.16 For the LSPS application, no noise exceedance is anticipated; therefore, the Project is expected to fully comply with the relevant noise standards.

4.6.2.17 Similar for Section 12A applications Y/YL-NSW/8 and Y/YL-NSW/9 (near the current Construction Sector Imported Labour Quarters), noise-conscious design and mitigation measures (such as acoustic windows and acoustic doors) will be implemented. With reference to the EA reports, 6.6 dB(A) and 6.4 dB(A) can be reduced using acoustic windows and acoustic doors.

4.6.2.18 For another Section 12A application Y/YL-NSW/7, its EA report stated that other than precautionary noise mitigation measures (e.g. single aspect design), baffle type acoustic windows/balconies will also be provided at dwellings with noise exceedance. The adjusted noise reduction ranged from 4.2 dB(A) to 6.9 dB(A) for acoustic windows applied at different room sizes and 4.2 dB(A) to 4.8 dB(A) for the acoustic doors.

4.6.2.19 For another Section 12A application Y/NE-KTS/18 of the proposed school at Hang Tau, its EA report stated that most noise sensitive uses, including classroom and clinic, will be air-conditioned and not rely on opened windows. For student and staff dormitory, their noise sensitive facades or location of openable windows would be designed facing away from the main noise sources, including the Project, as practicable as possible.

4.6.2.20 A 6 dB(A) noise reduction from at-receiver mitigation measures is assumed for the public housing developments at Sha Po according to the Preliminary Environmental Review (PER) under CEDD's CE8/2022 making reference from EPD's ProPECC PN 5/23. The proposed mitigation measures at the receiver for the residential towers, which will be further reviewed in the future EAS, have been acknowledge and committed for implementation by HKHA according to the PER.

4.6.2.21 HyD will inform the developers and relevant government departments of the abovementioned planned developments, that the design of Northern Metropolis Highway and its at-source mitigation measures were developed based on the layout, design, and at-receiver noise mitigation measures presented in their latest planning submissions and/or PER to ensure overall compliance with the road traffic noise standards. Should the developers or the respective departments subsequently modify their development layout or design, they should refer to this EIA when designing their at-receiver mitigation measures to maintain compliance with the road traffic noise standards.

4.6.2.22 The noise reduction performance of at-receiver mitigation measures for the existing and planned developments adopted in this EIA are summarized in **Table 4.14** below.

Table 4.14 Noise Reduction Performance of At-receiver Mitigation Measures for Existing and Planned Developments

Development	Proposed Noise Mitigation Measures and their Noise Reduction Performance
Park Yoho	Equipped with solid balcony with 2.0 dB(A) reduction
Application No. A/YL-NSW/293	Acoustic fins, enhanced acoustic windows (baffle type), and enhanced acoustic balconies (baffle type) Bedroom: -6.3 dB(A) Living room/ dining room: -7.0 dB(A)
Application No. Y/YLNSW/8	Acoustic windows and acoustic doors Acoustic window: -6.6 dB(A) for domestic portion and -8.8 for RCHE Acoustic Door: -6.4 dB(A)
Application No. Y/YLNSW/9	Acoustic windows and acoustic doors Acoustic Window: -6.6 dB(A) Acoustic Door: -6.4 dB(A)
Public Housing Developments at Sha Po	Acoustic Window: -6.0 dB(A)

4.6.3 Prediction and Evaluation of Road Traffic Noise Impact

Modelling Scenarios

4.6.3.1 Referring to the requirements of the EIA SB, the following scenarios were assessed in the EIA study.

- Without Project scenario at the assessment year (i.e. Year 2051);
- Unmitigated scenario at the assessment year (i.e. Year 2051);
- Mitigated scenario at the assessment year (i.e. Year 2051); and
- Prevailing scenario for indirect mitigated measures eligibility test (i.e. Year 2026).

Unmitigated Scenario

4.6.3.2 The predicted road traffic noise levels at each NAPs under unmitigated scenario are presented in **Table 4.15**. **Appendix 4.6** shows the details of the noise impacts at different levels of the NAPs under unmitigated scenario.

Table 4.15 Predicted Road Traffic Noise Impact at NAPs under Unmitigated Scenario

NSR Description	NAP	Use (Criteria, $L_{10\ 1hr}$, dB(A)) [1]	Predicted Noise Level, $L_{10\ 1hr}$ dB(A) [2]			Contribution by Project Road [2,4]	Noise Mitigation Measures by Project Road
			Other Road[3]	Project Road	Overall		
Existing							
Ko Po San Tsuen	KPST1, KPST2, KPST3	R (70)	67-71	44-58	67-71	0.2	N
HK Boy Scouts Association Yuen Long East District Centre	HKBSA	G/IC (70)	74	55-57	74	0.0-0.1	N
Au Tau Youth Centre	AT1	G/IC (70)	80	55	80	0.0	N
Ha Ko Po Tsuen	HKPT	R (70)	69	47	69	-	N
Riva	RV1, RV2	R (70)	40-43	<40	40-43	-	N
Crescent Green	CR1, CR2, CR3	R (70)	42-44	<40	42-44	-	N
Tung Shing Lei	TSL1a, TSL1b, TSL2, TSL3a, TSL3b	R (70)	51-70	51-70	54-73	3.2-3.9	Y
Park Yoho	PY1a, PY1b, PY2, PY3a, PY3b	R (70)	66-71	60-73	70-75	0.4-5.4	Y
Fung Kat Heung	FKHa FKHb	R (70)	68-69	70-72	73-74	3.4-5.4	Y
Shum Residence	SR1, SR2	R (70)	58-62	52-54	59-63	-	N
	SR3	W (65)	56-57	58-59	60-61	-	N

NSR Description	NAP	Use (Criteria, $L_{10\ 1hr}$, dB(A)) [1]	Predicted Noise Level, $L_{10\ 1hr}$ dB(A) [2]			Contribution by Project Road [2,4]	Noise Mitigation Measures by Project Road
			Other Road[3]	Project Road	Overall		
Mo Fan Heung	MFH1, MFH2, MFH3, MFH4, MFH5, MFH6, MFH7a, MFH7b, MFH7c, MFH8	R (70)	65-77	65-74	68-78	0.7-6.1	Y
Wing Kei Tsuen	WKT1, WKT2	R (70)	73-75	62-64	73-75	0.3-0.4	N
Elite Garden	EG1, EG2, EG3, EG4, EG5, EG6	R (70)	83-84	64-69	83-84	0.0-0.2	N
Pok Wai	PW1a, PW1b, PW1c, PW2a, PW2b, PW3a, PW3b, PW4a, PW4b, PW5, PW6	R (70)	59-83	60-69	63-83	0.0-1.9	Y
Merry Garden	MG1	R (70)	84	65	84	0.0-0.1	N
Long Ha	LHa, LHb	R (70)	57-73	64	65-73	0.0-0.5	N
Chan Ming Yam Tong	CMYT1a, CMYT1b, CMYT1c, CMYT2	R (70)	42-68	63-65	64-69	-	N
San Wai Tsuen	SWT1, SWT2a, SWT2b, SWT2c, SWT3a, SWT3b, SWT3c, SWT4	R (70)	<40-64	62-70	62-70	-	N
Ngau Tam Mei Village	NTMV1, NTMV2, NTMV3, NTMV4	R (70)	<40-50	62-77	62-77	22.7-26.6	Y
Cheung Lek	CL1, CL2a, CL2b	R (70)	<40	65-68	65-68	-	N

NSR Description	NAP	Use (Criteria, $L_{10\ 1hr}$, dB(A)) [1]	Predicted Noise Level, $L_{10\ 1hr}$ dB(A) [2]			Contribution by Project Road [2,4]	Noise Mitigation Measures by Project Road
			Other Road[3]	Project Road	Overall		
Ki Lun Tsuen South	KLTS1a, KLTS1b, KLTS2, KLTS3, KLTS4	R (70)	<40-51	60-73	60-73	33.2-36.1	Y
Ki Lun Tsuen North	KLTN1a, KLTN1b, KLTN1c, KLTN2a, KLTN2b, KLTN3a, KLTN3b, KLTN4a, KLTN4b	R (70)	53-63	55-69	57-69	-	N
Kwu Tung	KT1, KT2, KT3	R (70)	68-70	63-64	69-71	0.3-0.9	N
Kwu Tung North Multi-welfare Services Complex	KTN1	R (70)	70-71	58-59	70-71	0.2-0.3	N
Europa Garden	EPG1a, EPG1b, EPG2a, EPG2b, EPG3a, EPG3b, EPG4a, EPG4b, EPG5a, EPG5b, EPG6a, EPG6b	R (70)	57-78	<40-69	58-78	0.0-2.0	Y
Lady Ho Tung Welfare Centre Eco-Learn Institute	LHTWCa, LHTWCb	E (65)	60-68	<40	60-68	0.0	N
Valais	VL	R (70)	56-60	<40	56-60	-	N
Ma Tso Lung	MCLa, MCLb, MCLc	R (70)	<40	66-68	66-68	-	N
Planned							
Proposed Residential Development under Application No. A/YL-KTN/964	KTN964_1, KTN964_2, KTN964_3	R (70)	70-80	<40-56	70-80	0.0-0.1	N

NSR Description	NAP	Use (Criteria, $L_{10\ 1hr}$, dB(A)) [1]	Predicted Noise Level, $L_{10\ 1hr}$ dB(A) [2]			Contribution by Project Road [2,4]	Noise Mitigation Measures by Project Road
			Other Road[3]	Project Road	Overall		
Proposed Comprehensive Residential Development under Application No. A/YL-NSW/293	NSW293_1 NSW293_2, NSW293_3, NSW293_4, NSW293_5a, NSW293_5b, NSW293_5c, NSW293_6a, NSW293_6b, NSW293_6c, NSW293_7, NSW293_8, NSW293_9	R (70)	49-73	<40-76	54-78	3.2-5.8	Y
LSPS Application No. LSPS/002	LSPS1a, LSPS1b, LSPS2	R (70)	64-68	66-72	68-74	5.9-6.4	Y
Planned Public Housing Development and primary school at Sha Po under Agreement No. CE8/2022(CE)	SP1, SP2a, SP2b, SP3, SP4, SP5, SP6	R (70)	66-74	70-76	71-78	2.3-7.6	Y
	SP7	E (65)	58-60	51-57	59-61	-	N
Proposed Residential Development and School Site under Application No. Y/YL-NSW/9	NSW9_1a, NSW9_1b	E (65)	49-65	51-63	53-67	2.2-2.5	Y
	NSW9_2a, NSW9_2b, NSW9_2c, NSW9_2d, NSW9_2e, NSW9_2f, NSW9_3a, NSW9_3b, NSW9_3c, NSW9_3d, NSW9_3e, NSW9_3f	R (70)	62-72	64-78	64-79	3.9-11.5	Y
Proposed Residential Development under Application No. Y/YL-NSW/8	NSW8_1a, NSW8_1b, NSW8_1c, NSW8_1d, NSW8_1e, NSW8_1f, NSW8_2a, NSW8_2b,	R (70)	57-78	64-78	65-80	0.4-11.9	Y

NSR Description	NAP	Use (Criteria, $L_{10\ 1hr}$, dB(A)) [1]	Predicted Noise Level, $L_{10\ 1hr}$ dB(A) [2]			Contribution by Project Road [2,4]	Noise Mitigation Measures by Project Road
			Other Road[3]	Project Road	Overall		
	NSW8_2c, NSW8_2d, NSW8_3a, NSW8_3b						
Proposed Residential Development under Application No. Y/YL-NSW/7	NSW7_1, NSW7_2, NSW7_3,	R (70)	46-70	44-62	48-71	0.6	N
Proposed Social Welfare Facilities (Residential Care Home for the Elderly (RCHE)) under Application No. A/YL-NSW/348 and A/YL-NSW/349	NSW348, NSW349	R (70)	75	61-62	75	0.1-0.3	N
Proposed Residential Development under Application No. A/YL-NSW/314	NSW314_1, NSW314_2	R (70)	56-68	60-62	62-69	-	N
Proposed Residential Development, Staff Quarters and Student Hostels at Ngau Tam Mei New Development Area	NTM1, NTM2, NTM3, NTM4, NTM5, NTM6, NTM7, NTM8, NTM9, NTM10, NTM11, NTM12	R (70)	<40-70	62-79	62-79	3.7-77.4	Y
Proposed School at Kwu Tung South under	KTSIS1, KTSIS2	R (70)	<40	44-58	44-58	-	N

NSR Description	NAP	Use (Criteria, $L_{10\ 1hr}$, dB(A)) [1]	Predicted Noise Level, $L_{10\ 1hr}$ dB(A) [2]			Contribution by Project Road [2,4]	Noise Mitigation Measures by Project Road
			Other Road[3]	Project Road	Overall		
Application No. Y/NE-KTS/18							
Planned residential developments and schools at Kwu Tung North New Development Area	KTN2, KTN3	E (65)	48-68	<40-57	48-68	0.3-0.4	N
	KTN4, KTN5, KTN6, KTN7	R (70)	62-72	<40-61	62-73	0.0-0.4	N

Notes:
[1]: R: residential; E: educational institution; W: places of public worship.
[2]: AM Peak is presented.
[3]: Other road consists of existing road and committed road.
[4]: Contribution from Project Road for NSRs with overall noise level > noise criteria of 65/70dB(A).

4.6.4 Mitigation Measures

Direct Mitigation Measures

4.6.4.1 Referring to **Table 4.14** and **Appendix 4.6**, some NSRs will be exposed to road traffic noise levels exceeding the respective requirements. Direct noise mitigation measures listed below are proposed to mitigate the road traffic noise impact from the Project roads for those NSRs to a level that it:

- is not higher than the standard; and
- has no significant contribution (less than 1.0 dB(A)) to the cumulative noise level, if the overall noise level (i.e. noise from the Project Roads together with other existing, planned/committed roads) exceeds the noise standard.

4.6.4.2 Further to the LNRS provided for high speed road sections of the Project with speed limit of 80km/hr or above and for expressway (as discussed in **Section 4.5.2.7** and **Appendix 4.5**), LNRS (e.g. 6mm Polymer Modified Stone Mastic Asphalt (PMSMA6) or equivalent) has been adopted on slip roads of the Project as a mitigation measures. In addition to the use of LNRS, noise barriers have also been adopted and are proposed at locations as shown in **Figure 4.4**. For incremental approach, the noise barriers for protecting the planned NSRs will be implemented before the population intake of the planned developments. Summary of proposed at-source noise mitigation measures and their configuration at different stages of the Project is shown in **Table 4.16**. Locations of the direct mitigation measures are shown in **Figure 4.4**. Locations of mitigation measures for protecting existing NSRs are shown in **Figure 4.5**.

Table 4.16 Recommended At-source Noise Mitigation Measures

Location	Noise Mitigation Measures ID[1]	Type of Noise Mitigation Measures	Height above Road level (m)	Approximate Length (m)	NSRs to be Protected	When to Implement?
NM Highway near East and North of Tung Shing Lei	V01 (ST)	Vertical Noise Barrier	3	115	TSL1-2 (Existing)	Before commencement of NMH - ST
	V02 (TSW)	Vertical Noise Barrier	5	145	NSW293 (Planned)	Before population intake of the planned development and commencement of NMH - TSW
	V03 (TSW)	Vertical Noise Barrier	5	260	PY1 (Existing), NSW293 (Planned)	Before commencement of NMH - TSW
	C01 (ST)	Cantilevered noise barrier (absorptive)	5+3	210	TSL1, PY1 (Existing)	Before commencement of NMH - ST
	C02 (TSW)	Cantilevered noise barrier	7+3	200	NSW293_5-7 (Planned)	Before population intake of the planned development and commencement of NMH - TSW
	C03 (TSW)	Cantilevered noise barrier	5+3	205	LSPS1-2 (Planned)	Before population intake of the planned development and commencement of NMH - TSW
	C04 (TSW)	Cantilevered noise barrier	5+3	315	LSPS1-2 (Planned)	Before population intake of the planned development and commencement of NMH - TSW

Location	Noise Mitigation Measures ID[1]	Type of Noise Mitigation Measures	Height above Road level (m)	Approximate Length (m)	NSRs to be Protected	When to Implement?
NM Highway near Kam Tin River	V04 (ST)	Vertical Noise Barrier	3	320	LSPS1-2 (Planned)	Before population intake of the planned development
	V05 (ST)	Vertical Noise Barrier	5	200	PY1-3 (Existing)	Before commencement of NMH - ST
	V06 (TSW)	Vertical Noise Barrier	5	290	LSPS1-2 (Planned)	Before population intake of the planned development and commencement of NMH - TSW
	C05 (TSW)	Cantilevered noise barrier	5+3	450	PY1-3 (Existing); SP1-3 (Planned)	Before population intake of the planned development and commencement of NMH - TSW
	LNRS1&3 (ST) LNRS2&4 (TSW)	Low Noise Road Surfacing provided on the slip roads	-	2945	PY1-3 (Existing); NSW293_5-7, LSPS1-2, SP1-3, NSW9_1-3 (Planned)	Before commencement of NMH – ST and NMH - TSW
	V07 (TSW)	Vertical Noise Barrier	5	200	FKH (Existing); SP1-3 (Planned)	Before commencement of NMH - TSW
NM Highway near San Tin Highway	V08 (ST)	Vertical Noise Barrier	5	120	SP1-3 (Planned)	Before population intake of the planned development
	V09 (ST)	Vertical Noise Barrier. (absorptive)	3	375	NSW9_1-3, NSW8_1-2 (Planned)	Before population intake of the planned development

Location	Noise Mitigation Measures ID[1]	Type of Noise Mitigation Measures	Height above Road level (m)	Approximate Length (m)	NSRs to be Protected	When to Implement?
	V10[2] (ST)	Vertical Noise Barrier (absorptive)	5	270	MFH1-2 (Existing); SP1-6 (Planned)	Before commencement of NMH - ST
	C06 (ST)	Cantilevered noise barrier	5+3	250	NSW9_1-3, (Planned)	Before population intake of the planned development
	C07 (ST)	Cantilevered noise barrier	5+3	145	NSW9_1-3 (Planned)	Before population intake of the planned development
	C08 (ST)	Cantilevered noise barrier	5+3	245	SP1-3 (Planned)	Before population intake of the planned development
	C09 (ST)	Cantilevered noise barrier	7+3	470	NSW9_1-3, NSW8_1-3 (Planned)	Before population intake of the planned development
	C10 (ST)	Cantilevered noise barrier (absorptive)	7+3	215	SP1-6 (Planned)	Before population intake of the planned development
	C11 (ST)	Cantilevered noise barrier (absorptive)	7+3	380	MFH3-8 (Existing); SP4-6 (Planned)	Before commencement of NMH - ST
	C12 [3] (ST)	Cantilevered noise barrier	5+3	140	MFH1-2 (Existing); SP4-6 (Planned)	Before commencement of NMH - ST
NM Highway near Pok Wai	V11 (ST)	Vertical Noise Barrier	3.5	165	PW1-3 (Existing)	Before commencement of NMH - ST
	V12 (ST)	Vertical Noise Barrier	3	130	NTMV4 (Existing)	Before commencement of NMH - ST

Location	Noise Mitigation Measures ID[1]	Type of Noise Mitigation Measures	Height above Road level (m)	Approximate Length (m)	NSRs to be Protected	When to Implement?
NM Highway near Ngau Tam Mei	C13 [4] (ST)	Cantilevered noise barrier (absorptive)	7+4	900	NTM5-12 (Planned)	Before population intake of the planned development
	C14 [4] (ST)	Cantilevered noise barrier (absorptive)	7+4	630	NTM5-12 (Planned)	Before population intake of the planned development
NM Highway near Ki Lung Tsuen	V13 (ST)	Vertical Noise Barrier	2	200	KLTS3-4 (Existing)	Before commencement of NMH - ST
NM Highway near Fanling Highway	C15 (ST)	Cantilevered noise barrier	5+3	70	EPG6 (Existing)	Before commencement of NMH - ST
	C16 [5] (ST)	Cantilevered noise barrier (absorptive)	7+3	95	EPG6 (Existing)	Before commencement of NMH - ST

Note:
[1]: Proposed mitigation measures to be implemented under NM Highway San Tin section (ST) or NM Highway Tin Shui Wai section (TSW).
[2]: For the section of the existing 3m noise barrier NB205 along San Tin Highway that overlaps with the proposed 5m barrier (V10), the overlapping portion will be removed, and the remaining section will be repositioned to align with the San Tin Highway.
[3]: As shown in **Figure 4.5.2**, the extent of noise barrier would be in form of 3m vertical noise barrier for existing sensitive uses, an increment approach would be adopted and the noise barrier would be upgrade to C12 cantilevered noise barrier for planned sensitive uses before the population intake.
[4]: Based on NTM EIA, the noise exceedance at planned staff quarters and student hostels at Site G.12 should be mitigated under the NMH-ST EIA study. Under this EIA, cantilevered noise barriers (C13 and C14) have been recommended to mitigate at-source, while these mitigation measures are indicative only and are subject to further study together with other direct at-receiver mitigation measures by future developers based on their final development layout.
[5]: Part of the proposed KTN-NB26 will be removed to provide the slip road for NM Highway and it is re-provisioned as C16 under NM Highway.

4.6.4.3 Apart from the at-source noise mitigation measures, at-receiver mitigation measures e.g. acoustic fins, noise-conscious design and acoustic windows/balconies have been proposed at the planned residential sites as stated in **Table 4.17**. These mitigation measures are indicative only and are subject to further study together with other direct at-receiver mitigation measures by future developers based on their final development layout. The design and respective noise reduction performance of the acoustic windows/balconies should make reference to ProPECC PN 5/23, and should comply with the Building Department's Practice Note for Authorized Persons, Registered Structural Engineers and Registered Geotechnical Engineers (No. APP-130) Lighting and Ventilation Requirements – Performance-based Approach to achieve respective ventilation requirements.

4.6.4.4 To ensure the effectiveness of the direct at-receiver mitigation measures, noise impact assessment (NIA) or environmental assessment study (EAS) or environmental assessments under the appropriate mechanism shall be undertaken by the future developers. The NIA/EAS/environmental submissions shall be conducted in accordance with Chapter 9 of the Hong Kong Planning Standards and Guidelines (HKPSG), and explore direct at-receiver mitigation measures to ensure compliance with the relevant noise criteria. HyD will follow up with the relevant authorities to have the requirements of NIA/EAS/environmental submissions incorporated in respective lease documents, planning approval conditions, or through other appropriate mechanism.

4.6.4.5 With the implementation of the recommended at-source noise mitigation measures, including LNRS, cantilevered and vertical noise barriers and at-receiver noise mitigation measures, no adverse road traffic noise impact from the Project Road is anticipated. While the overall traffic noise at some representative NSRs still exceed the standard, the noise contribution from the Project Road is less than 1.0 dB(A), and those exceedances are attributable to other roads. The predicted overall noise levels of all the NSRs are summarized in **Table 4.17** and presented in **Appendix 4.7**.

Table 4.17 Predicted Road Traffic Noise Impact at NAPs under Mitigated Scenario

NSR Description	NAP	Use (Criteria, $L_{10\ 1hr}$, dB(A)) [1]	Predicted Noise Level, $L_{10\ 1hr}$ dB(A) [2]			Contribution by Project Road [4]	Adopted Noise Reduction Performance of At-receiver Mitigation Measures, dB(A)	Overall Noise Level with At-receiver base Noise Mitigation Measures, $L_{10\ 1hr}$ dB(A)	Whether Noise Mitigation Measures on "Project Roads" are Required? (Y/N)
			Other Road [3]	Project Road	Overall				
Existing NSRs									
Tung Shing Lei	TSL1a, TSL1b, TSL2, TSL3a, TSL3b	R (70)	51-68	50-65	53-70	-	N/A	N/A	N
Park Yoho	PY1a, PY1b, PY2, PY3a, PY3b	R (70)	66-71	53-66	67-72	0.3-1.0	2.0	70	N
Fung Kat Heung	FKHa FKHb	R (70)	68-69	64-65	70	-	N/A	N/A	N
Mo Fan Heung	MFH1, MFH2, MFH3, MFH4, MFH5, MFH6, MFH7a, MFH7b, MFH7c, MFH8	R (70)	64-76	58-65	66-77	0.2-0.5	N/A	N/A	N
Pok Wai	PW1a, PW1b, PW1c, PW2a, PW2b, PW3a, PW3b, PW4a, PW4b, PW5, PW6	R (70)	59-83	58-65	63-83	0.0-0.8	N/A	N/A	N
Ngau Tam Mei Village	NTMV1, NTMV2, NTMV3, NTMV4	R (70)	<40	54-69	54-69	-	N/A	N/A	N
Ki Lun Tsuen South	KLTS1a, KLTS1b, KLTS2, KLTS3, KLTS4	R (70)	<40-51	60-69	60-69	-	N/A	N/A	N

NSR Description	NAP	Use (Criteria, $L_{10\ 1hr}$, dB(A)) [1]	Predicted Noise Level, $L_{10\ 1hr}$ dB(A) [2]			Contribution by Project Road [4]	Adopted Noise Reduction Performance of At-receiver Mitigation Measures, dB(A)	Overall Noise Level with At-receiver base Noise Mitigation Measures, $L_{10\ 1hr}$ dB(A)	Whether Noise Mitigation Measures on "Project Roads" are Required? (Y/N)
			Other Road [3]	Project Road	Overall				
Europa Garden	EPG1a, EPG1b, EPG2a, EPG2b, EPG3a, EPG3b, EPG4a, EPG4b, EPG5a, EPG5b, EPG6a, EPG6b	R (70)	57-72	<40-64	57-72	0.0-0.8	N/A	N/A	N
Planned NSRs									
Proposed Comprehensive Residential Development under Application No. A/YL-NSW/293	NSW293_1NSW293_2, NSW293_3, NSW293_4, NSW293_5a, NSW293_5b, NSW293_5c, NSW293_6a, NSW293_6b, NSW293_6c, NSW293_7, NSW293_8, NSW293_9	R (70)	54-73	<40-74	50-77	2.5-6.4	6.3	70	N
LSPS Application No. LSPS/002	LSPS1a, LSPS1b, LSPS2	R (70)	64-67	58-68	65-70	-	N/A	N/A	N
Planned Public Housing Development and primary school at Sha Po under	SP1, SP2a, SP2b, SP3, SP4, SP5, SP6	R (70)	64-74	61-75	61-76	0.4-6.9	6.0	70	N

NSR Description	NAP	Use (Criteria, $L_{10\ 1hr}$, dB(A)) [1]	Predicted Noise Level, $L_{10\ 1hr}$ dB(A) [2]			Contribution by Project Road [4]	Adopted Noise Reduction Performance of At-receiver Mitigation Measures, dB(A)	Overall Noise Level with At-receiver base Noise Mitigation Measures, $L_{10\ 1hr}$ dB(A)	Whether Noise Mitigation Measures on "Project Roads" are Required? (Y/N)
			Other Road [3]	Project Road	Overall				
Agreement No. CE8/2022(CE)									
Proposed Residential Development and School Site under Application No. Y/YL-NSW/9	NSW9_1a, NSW9_1b	E (65)	49-64	49-59	52-65	-	N/A	N/A	N
	NSW9_2a, NSW9_2b, NSW9_2c, NSW9_2d, NSW9_2e, NSW9_2f, NSW9_3a, NSW9_3b, NSW9_3c, NSW9_3d, NSW9_3e, NSW9_3f	R (70)	51-72	51-75	59-77	0.4-7.7	6.4	70	N
Proposed Residential Development under Application No. Y/YL-NSW/8	NSW8_1a, NSW8_1b, NSW8_1c, NSW8_1d, NSW8_1e, NSW8_1f, NSW8_2a, NSW8_2b, NSW8_2c, NSW8_2d, NSW8_3a, NSW8_3b	R (70)	54-78	57-75	59-79	0.0-9.5	6.4	70	N
Proposed Residential Development, Staff Quarters and Student Hostels at Ngau Tam Mei New	NTM1, NTM2, NTM3, NTM4, NTM5, NTM6, NTM7, NTM8, NTM9, NTM10, NTM11, NTM12	R (70)	<40-70	55-74	55-74	3.7-11.1	6.0	68	N

NSR Description	NAP	Use (Criteria, $L_{10\ 1hr}$, dB(A)) [1]	Predicted Noise Level, $L_{10\ 1hr}$ dB(A) [2]			Contribution by Project Road [4]	Adopted Noise Reduction Performance of At-receiver Mitigation Measures, dB(A)	Overall Noise Level with At-receiver base Noise Mitigation Measures, $L_{10\ 1hr}$ dB(A)	Whether Noise Mitigation Measures on "Project Roads" are Required? (Y/N)
			Other Road [3]	Project Road	Overall				
Development Area									

Notes:
[1]: R: residential; E: educational institution
[2]: AM Peak is presented.
[3]: Other road consists of existing road and committed road.
[4]: Contribution from Project Road for NSRs with overall noise level > noise criteria of 70dB(A).

4.6.5 Indirect Mitigation Measures

4.6.5.1 According to Section 4.8 of EIAO GN No. 12/2023, the testing criteria for consideration of Indirect Mitigation Measures are set out as below:

- the predicted overall noise level from the road project together with other road traffic noise in the vicinity must be above the standard laid down in Annex 5 of the EIAO-TM;
- the predicted overall road traffic noise level is at least 1.0 dB(A) more than the prevailing road traffic noise level, i.e. the total road traffic noise level existing before the works to construct the road were commenced; and
- the contribution to the increase in the predicted overall road traffic noise level from the road project must be at least 1.0 dB(A).

4.6.5.2 The assessment results of the without project scenario at Year 2051 and prevailing scenario at Year 2026 are shown in **Appendix 4.6** and **Appendix 4.7**. After the implementation of direct mitigation measures, those NSRs with overall road traffic noise levels exceeding the relevant noise criteria, the contribution from "Project Road" to the overall road traffic noise level is less than 1.0dB(A). Therefore, all those NSRs would not satisfy the eligibility assessment criteria in accordance with **Section 4.6.2.5**.

4.6.6 Residual Impact

4.6.6.1 With the implementation of all the proposed direct noise mitigation measures, including at-source mitigation measures by the Project and at-receiver mitigation measures at planned developments, the Project road noise contributions to the overall noise levels at all NSRs would be less than 1.0 dB(A) and the Project road noise levels would all be below the relevant noise criteria. Adverse road traffic noise impact on existing and planned NSRs is not anticipated.

4.6.7 Evaluation of Protected and Benefitted Noise Sensitive Uses with the Noise Mitigation Measures

4.6.7.1 To study the noise performance of the Project, traffic noise levels at the residential, schools and other noise sensitive uses including place of worship which have a direct line of sight to the Project have been predicted. The numbers of dwellings, classrooms and other noise sensitive uses that would be benefited from and be protected by the provision of noise mitigation measures have been calculated. The definition of "exposed", "benefited" and "protected" noise sensitive uses are defined as follow:

- Exposed – Noise sensitive uses with unmitigated noise level greater than relevant noise criteria
- Benefited – Exposed noise sensitive uses with a noise reduction of 1.0 dB(A) or greater in overall noise level with the noise mitigation measures in place
- Protected – Exposed noise sensitive uses with an overall noise level not greater than relevant noise criteria with the noise mitigation measures in place

4.6.7.2 Number of dwellings and classrooms that would be benefited from and be protected by the provision of noise mitigation measures will be identified for existing residential and schools respectively. Moreover, the planned noise sensitive uses do not have detailed numbers of dwellings/ classrooms, the benefitted/ protected will be identified

as the number of floors. Results of existing and planned noise sensitive uses are presented in tables below.

Table 4.18 Summary of Protected and Benefited Noise Sensitive Uses (Existing)

Noise Sensitive Uses	Total No. of Dwellings/ Classrooms/ Noise Sensitive Uses Considered	Unmitigated Scenario	Mitigated Scenario		
		No. of Exposed Dwellings/ Classrooms/ Noise Sensitive Uses	No. of Exposed Dwellings/ Classrooms/ Noise Sensitive Uses	Protected Dwellings/ Classrooms/ Noise Sensitive Uses	Benefitted Dwellings/ Classrooms/ Noise Sensitive Uses
Dwellings	972	854	698	158	164
Classrooms	1	1	1	0	0
Government, Institution or Community (G/IC)	11	8	8	0	0

Table 4.19 Summary of Protected and Benefited Noise Sensitive Uses (Planned)

Noise Sensitive Uses	Total No. of Floors of Dwellings/ Classrooms/ Noise Sensitive Uses Considered	Unmitigated Scenario	Mitigated Scenario		
		No. of Exposed Floors of Dwellings/ Classrooms/ Noise Sensitive Uses	No. of Exposed Floors of Dwellings/ Classrooms/ Noise Sensitive Uses	Protected Floors of Dwellings/ Classrooms/ Noise Sensitive Uses	Benefitted Floors of Dwellings/ Classrooms/ Noise Sensitive Uses
Dwellings	1315	1117	984	133	617
Classrooms	15	10	6	4	7
Government, Institution or Community (G/IC)	0	0	0	0	0

4.7 Fixed Noise Sources Impact Assessment

4.7.1 Assessment Methodology

4.7.1.1 As set out in Appendix C Clause 3.2.3 of the EIA SB, a qualitative assessment is required to identify the fixed noise sources/ temporary industrial sources, and proposing corresponding direct mitigation measures to be adopted during operational phase to demonstrate that no adverse fixed noise sources impact will be associated with the Project. A summary of key steps for this qualitative fixed noise sources noise assessment that has conducted is:

- Determine the assessment area from Project Road and highway / tunnel operation and maintenance facilities;
- Summarize the fixed noise sources;
- Identify and locate representative NSRs that may be affected by the noise sources;
- Determine the noise criteria for both daytime and nighttime;
- Evaluate the potential qualitative impact at the NSRs; and
- Examine and recommend all practical mitigation measures such as quiet plant, silencer, enclosure, etc., to alleviate any potential noise impacts as much as practicable.

4.7.2 Identification of Fixed Noise Sources Impact

Inventory of Noise Sources

4.7.2.1 According to current design, there is one administration building near Ngau Tam Mei Development Area and some portal plant building near the tunnel portals proposed and summarized in **Table 4.20** below. Potential fixed noise impacts arising from the administration building and portal plant building would be caused by operation of the ventilation fans. The location of the potential fixed noise source during operational phase are shown in **Figure 4.6**.

Table 4.20 Summary of Location of Portal Plant Buildings and Administration Building

Location	Fixed Noise Source
South of Ngau Tam Mei Development Area	(F1) Proposed Administration Building
South of Tunnel A – Ngau Tam Shan Tunnel	(F2) Portal Plant Building (PPBA1)
North of Tunnel A – Ngau Tam Shan Tunnel	(F3) Portal Plant Building (PPBA2)
Pang Loon Tei Underpass	(F4) Proposed Minor Control Point (MCP2)
South of Tunnel B – Ki Lun Shan Tunnel	(F5) Portal Plant Building (PPBB1)

Location	Fixed Noise Source
North of Tunnel C – Kwun Tung South Underpass	(F6) Portal Plant Room (PPRC1)

Identification of Assessment Area and NSR

4.7.2.2 According to Clause 3.1.1(a) of Appendix C of the EIA Study Brief, the assessment area for fixed noise source impact includes an area within 300m from the boundary of the Project and its associated works of the Project. Representative NSRs locations that would be affected by the fixed noise sources have been identified and presented in **Table 4.10** and the noise criteria are summarised in the **Table 4.21** below. Locations of NAPs for fixed noise sources impact assessment are shown in **Figure 4.6**.

4.7.2.3 Annex 5 of the EIAO-TM has specified the following requirements for the planned fixed noise sources, whichever is more stringent.

- 5dB(A) below the appropriate ANLs shown in Table 2 of the IND-TM; or
- the prevailing background noise levels.

Table 4.21 Representative NAPs for Fixed Noise Source Impact Assessment

Fixed Noise Source ID.	NAP	Measurement Location	Nearest Distance between Fixed Noise Source and NAP (m)	Type of Area	Influencing Factor	Area Sensitivity Rating	ANL-5, dB(A) [1]	Prevailing Background Noise Level, dB(A) [1]	Noise Criteria, dB(A) [1]
F1	NTM3	Ngau Tam Mei	250	Area other than above [2]	-	B [2]	60/50	47/42	47/42
F2	NTM11	Ngau Tam Mei	130	Area other than above [2]	-	B [2]	60/50	41/38	41/38
F3	CL1	Ki Lun Tsuen South	220	Village Type	-	A	55/45	50/48	50/45
F4	KLTS4	Ki Lun Tsuen South	235	Village Type	-	A	55/45	50/48	50/45
F5	KLTN4	Ki Lun Tsuen North	150	Village Type	-	A	55/45	49/48	49/45
F6	KT1	Ki Lun Tsuen North	225	Village Type	Fanling Highway	B	60/50	49/48	49/45

Notes:
[1]: Daytime (0700 to 1900 hours) and Evening (1900 to 2300 hours) / Night (2300 to 0700 hours).
[2]: Reference from Development at Ngau Tam Mei Area EIA (AEIAR-271/2025).

4.7.3 Evaluation of Fixed Noise Sources Impact

4.7.3.1 One administration building and several portal plant buildings/ rooms are proposed and are the key fixed noise sources for the Project. NSRs in the vicinity of these fixed noise sources include the planned residential development and student hostels at Ngau Tam Mei and village houses near Cheung Lek, Ki Lun Tsuen and Kwu Tung as identified in **Figure 4.6**.

4.7.3.2 The distance between the proposed fixed noise sources and the nearest representative noise sensitive receivers are in between 130m to 250m. It is anticipated that, with a combination of quieter plant, installation of suitable sound attenuators, suitable orientation of the key louvers, etc., adverse fixed noise impact is not anticipated. The detailed designer / contractor of the ventilation building shall consider all the contemporary circumstances including but not limited to the ventilation building for the Project when designing the noise mitigation measures, and submit the Fixed Noise Source Management Plan (discussed in **Section 4.7.6**) for approval before commencement of construction, when designing the noise mitigation measures.

4.7.4 Mitigation of Fixed Noise Sources Impact

4.7.4.1 Possible mitigation measures with reference to EPD's "Good Practices on the Control of Noise from Electrical & Mechanical Systems" could be considered the fixed noise sources, for example:

- Quieter equipment;
- Silencer;
- Acoustic louvers;
- Barrier; and
- Enclosure, etc.

4.7.4.2 The detailed design should incorporate the following good practice in order to minimize the nuisance on the neighboring NSRs. In case the contractor would change the design and locations of the vents, they would need to comply with the legislative impacts at the receivers.

- Louvres should be orientated away from adjacent NSRs, preferably onto main roads which are less sensitive; and
- The façade for these ventilation shafts should have adequate sound insulation properties to minimise the noise emanating through the building fabric.

4.7.5 Residual Impact

4.7.5.1 With the implementation of mitigation measures, no residual fixed noise impact is anticipated.

4.7.6 Fixed Noise Sources Management Plan

4.7.6.1 As set out in Appendix C Clause 3.4 of the EIA SB and as detailed in EIAO GN No. 16/2023, a Fixed Noise Source Management Plan (FNMP) containing the quantitative fixed noise sources impact assessment, covering planned and existing fixed noise sources, noise mitigation measures and fixed noise sources impact monitoring and audit programme will be submitted to the EPD for agreement with reference to the updated and identified plant inventories once available and in any case prior to

issuance of tender and before commencement of construction of the proposed fixed noise sources.

4.7.6.2 The FNMP will include an implementation schedule to clearly list out the mitigation measures, the implementation party, locations and timing of implementation. Mitigation measures recommended and requirement specified in the FNMP shall be fully implemented by the Contactor.

4.8 Environmental Monitoring and Auditing

4.8.1 Construction Noise

4.8.1.1 Noise monitoring should be carried out as part of the EM&A programme for the construction phase of the Project to check compliance with the construction noise criteria.

4.8.1.2 A CNMP should evaluate the potential construction noise impacts based on the latest construction programme and construction plant inventory, and to assess the effectiveness and practicality of all proposed noise mitigation measures. As stipulated in s.6.2 of EIAO GN No. 9/2023, the CNMP usually needs to be submitted to the Director no later than 2 months before the tender invitation of the Project, if any, and before commencement of the construction of the Project implementation. If there is any change to the construction noise mitigation measures and/or plant inventory recommended in the submitted CNMP, an updated CNMP should be submitted to the Director, no later than one month before the implementation of any of such change. The implementation of the mitigation measures recommended in CNMP should also be audited as part of the EM&A programme. Details of the EM&A requirements are provided in a stand-alone EM&A Manual.

4.8.2 Road Traffic Noise

4.8.2.1 Road traffic noise monitoring should be carried out during operational phase at representative NSRs located in the vicinity of the recommended direct mitigation measures. Road traffic noise levels should be monitored at representative existing NSRs, which are in the vicinity of the recommended direct mitigation measures, during the first year after road opening. The purpose of the monitoring is to ascertain that the recommended mitigation measures are effective in reducing the noise levels. Details of the programme are provided in the EM&A Manual.

4.8.3 Fixed Noise

4.8.3.1 A FNMP should evaluate the potential fixed noise sources impacts and to assess the effectiveness and practicality of all proposed noise mitigation measures. The FNMP would be prepared prior to issuance of tender and before commencement of construction of the Project. Fixed noise audit should be carried out for the proposed fixed noise sources during the testing and commissioning stage to verify the compliance of the EIAO-TM criteria.

4.9 Conclusion

4.9.1 Construction Noise

4.9.1.1 Construction noise impact of the Project has been evaluated according to the requirements in the EIA Study Brief. With the adoption of proposed noise mitigation

measures, including the use of QPME, use of quieter construction method/equipment, use of higher surface density site-hoarding, use of noise barrier / enclosure, etc., adverse noise impact arising from construction works of the Project is not anticipated. Nevertheless, a CNMP, which contains a quantitative construction noise impact assessment, mitigation measures and monitoring and audit programme, will need to be submitted to the Director no later than 2 months before the tender invitation of the Project, if any, and before commencement of the construction of the Project. If there is any change to the construction noise mitigation measures and/or plant inventory recommended in the submitted CNMP, an updated CNMP should be submitted to the Director, no later than one month before the implementation of any of such change.

4.9.2 Road Traffic Noise

4.9.2.1 A road traffic noise assessment has been conducted according to the requirements in EIA Study Brief. Mitigation measures, such as LNRS, vertical and cantilevered type noise barriers as well as at-receiver mitigation measures (acoustic window/door/balcony) have been recommended to be implemented within the Project Road extent to fulfil the respective requirements.

4.9.3 Fixed Noise

4.9.3.1 Fixed noise sources impact of the Project has been evaluated according to the requirements in EIA Study Brief. With the adoption of proposed noise mitigation measures, including the use of quieter equipment, silencer, barrier, enclosure, etc., adverse noise impact arising from fixed noise sources is not anticipated. Nevertheless, a FNMP, which contains a quantitative fixed noise sources impact assessment, mitigation measures and monitoring and audit programme, should be submitted prior to issuance of the tender and before commencement of construction of the proposed fixed noise sources.